Introduction to IT

Lecture 1

Outlines

- Data, Information & Process
- Knowledge
- Characteristics of Valuable Information
- Knowledge workers & Knowledge management
- What is an information system
- Information Technology
- IS Versace IT

Data, Information & Process

- **Definition: Data** consists of raw facts.
- such as an employee number, total hours worked in a week, an inventory part number.
- Data represents real-world things.

Types of data

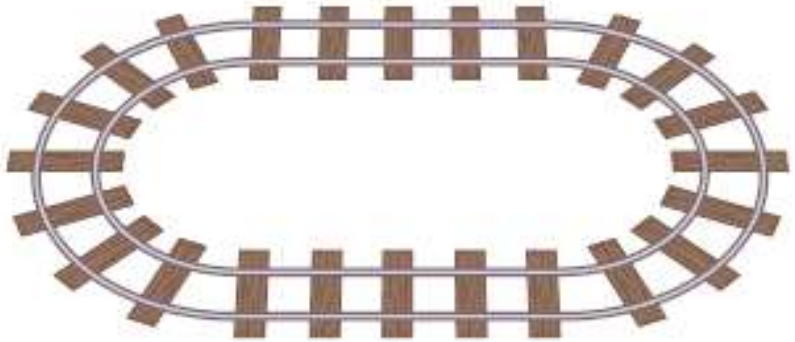
Data	Represented by
Alphanumeric data	Numbers, letters, and other characters
Image data	Graphic images and pictures
Audio data	Sounds, noises, or tones
Video data	Moving images or pictures

Data, Information & Process

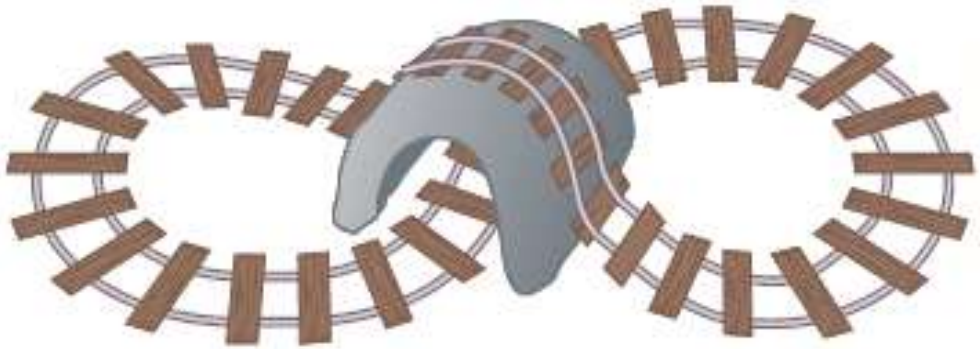
- **Information** is a collection of data organized and processed so that it has additional value beyond the value of the individual facts.
- **Process:** a set of logically related tasks performed to achieve a defined outcome.

Data, Information & Process

(a)



(b)



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Data, Information & Process

- Consider data as pieces of railroad track in a model railroad kit.
- Each piece of track has limited inherent value as a single object.
- However, if you define a relationship among the pieces of the track, they gain value.
- By arranging the pieces in a certain way, a railroad layout begins to emerge.

Data, Information & Process

- The type of information created depends on the relationships defined among existing data.
- For example, you could rearrange the pieces of track to form different layouts.
- Adding new or different data means you can redefine relationships and create new information.
- For instance, adding new pieces to the track can greatly increase the value.

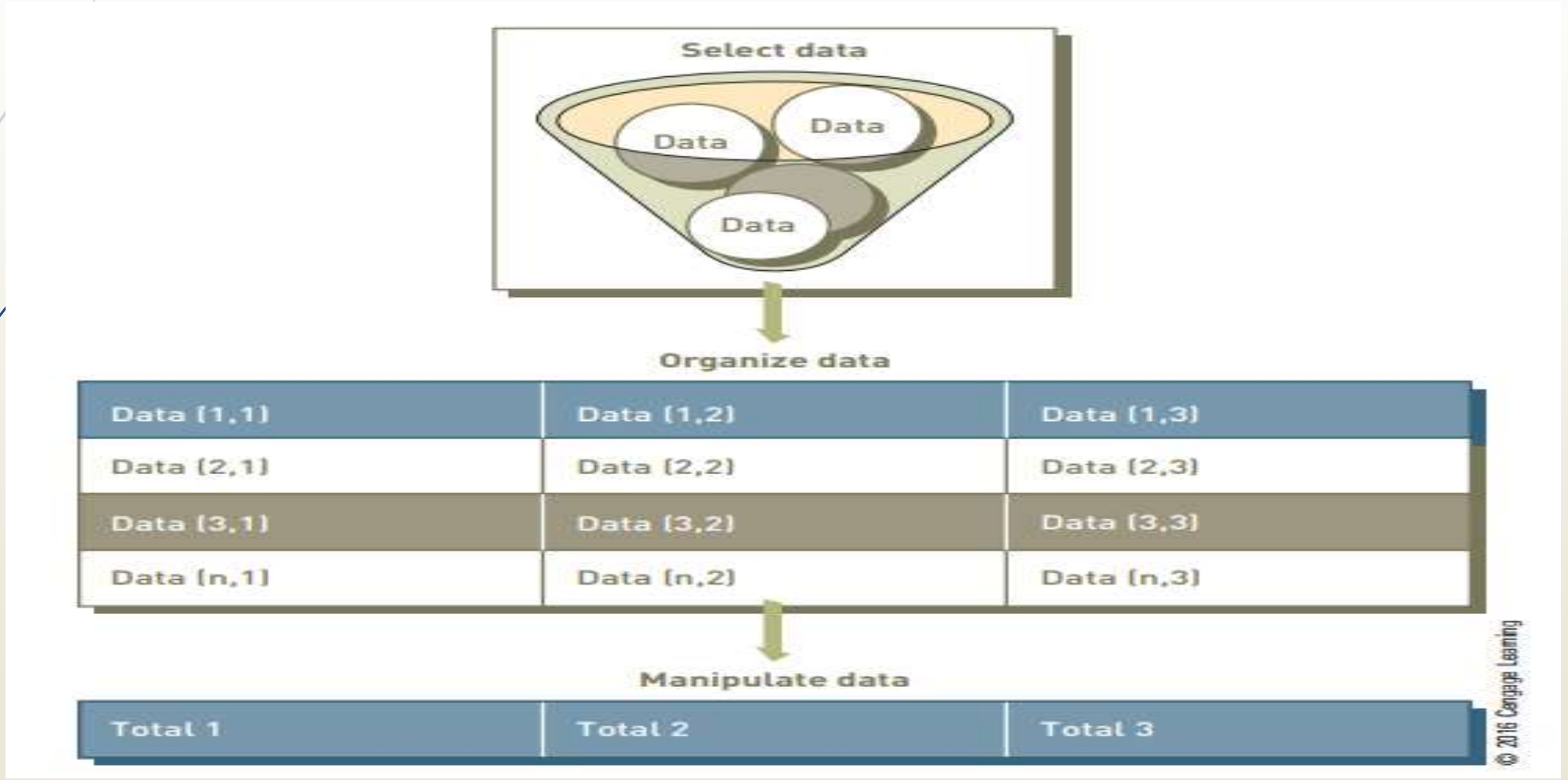
Knowledge

- The process of defining relationships among data to create useful information requires **knowledge**.
- **Knowledge** is the awareness and understanding of a set of information and the ways that information can be made useful to support a specific task or reach a decision.

Process of transforming data into information

- Transforming data into information starts by selecting data, then organizing it, and finally manipulating the data.

Process of transforming data into information



Characteristics of Valuable Information

- The value of information is directly linked to how it helps decision makers achieve their organization's goals.
- Valuable information can help people in their organizations perform tasks more **efficiently** and **effectively**.

Characteristics of Valuable Information

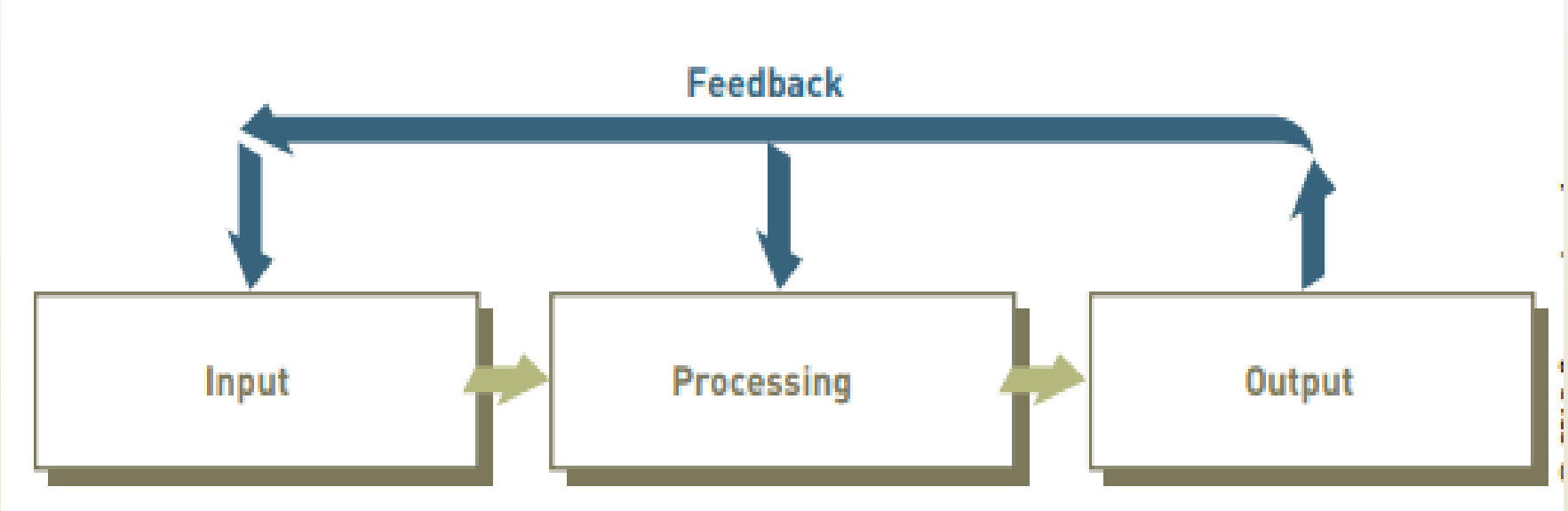
Characteristics	Definitions
Accessible	Information should be easily accessible by authorized users.
Accurate	Accurate information is error free.
Complete	Complete information contains all the important facts.
Flexible	Flexible information can be used for a variety of purposes.
Reliable	Reliable information can be trusted by users.
Secure	Information should be secure from access by unauthorized users.
Simple	Information should be simple, not complex.
Timely	Timely information is delivered when it is needed.
Verifiable	This means that you can check it to make sure it is correct

Knowledge workers & Knowledge management

- **Knowledge workers** (**KWs**) are people who create, use, and disseminate knowledge and are usually professionals in science, engineering, business, and other areas.
- **Knowledge management** is a strategy by which an organization determinedly and systematically gathers, organizes, stores, analyzes, and shares its collective knowledge and experience.

What is an Information System

- is a set of interrelated components that collect (**input**), manipulate (**processing**), store, and disseminate (**output**) data and information and provide a feedback mechanism to meet an objective.



What is an Information System

- In IS, **Input** is the activity of gathering and capturing raw data.
- In IS, **Processing** means converting or transforming data into useful outputs.
- Processing can be done **manually** or with **computer assistance**.

What is an Information System

- In IS, **Output** involves producing useful information, usually in the form of documents and reports.
- In IS, **feedback** is information from the system that is used to make changes to input or processing activities.
- In addition to feedback, a computer system can predict future events to avoid problems. This concept, often called **forecasting**

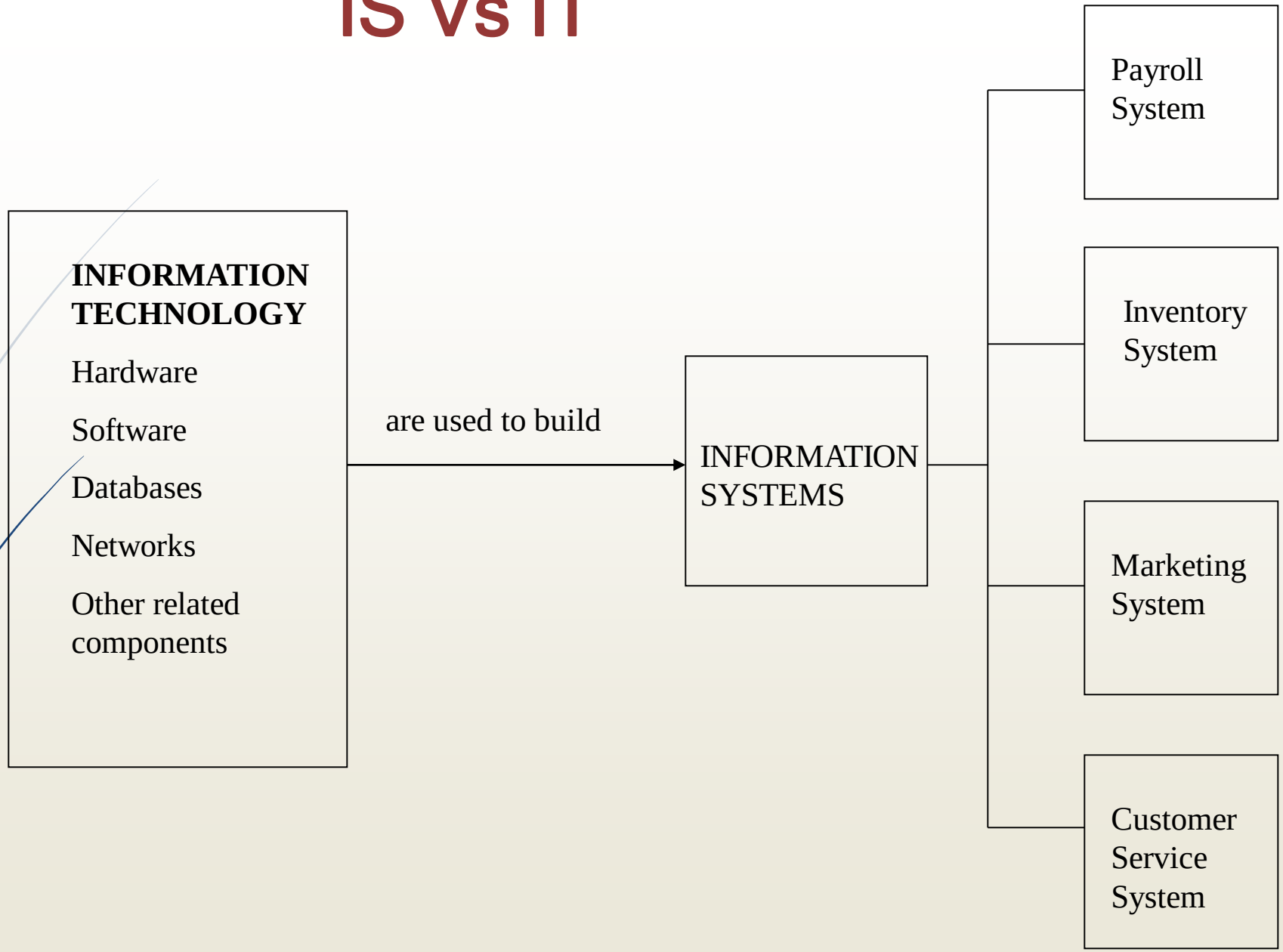
Information Technology

- **Information Technology** (IT) is the use of any computers, storage, networking and other physical devices, infrastructure and processes to create, process, store, secure and exchange all forms of electronic data.
- IT is used in the context of enterprise operations as opposed to personal or entertainment technologies. The commercial use of IT encompasses both computer technology and telephony.

Information Technology

- IT includes several layers of physical equipment (**hardware**), virtualization and management or automation tools, operating systems and applications (**software**) used to perform essential functions.

IS Vs IT


INFORMATION TECHNOLOGY
Hardware
Software
Databases
Networks
Other related components
are used to build
INFORMATION SYSTEMS
Payroll System
Inventory System
Marketing System
Customer Service System

Thanks !

Introduction to Information technology

Lecture 2

Outlines

- CBIS
- Components of CBIS

COMPUTER-BASED INFORMATION SYSTEMS

- **(CBIS):** A single set of hardware, software, databases, telecommunications, people, and procedures that are configured to collect, manipulate, store, and process data into information.
- Also called technology infrastructure
- Information technology (**IT**) refers to hardware, software, databases, and telecommunications.

COMPUTER-BASED INFORMATION SYSTEMS



Databases



Procedures



Hardware



Software



Networks

Components of a CBIS

- **1) Hardware**:
- consists of computer equipment used to perform input, processing, storage, and output activities.
- Processing devices include computer chips that contain the central processing unit and main memory.
- Advances in **chip** design allow faster speeds, less power consumption, and larger storage capacity, which means the computer can perform more efficiently.

Components of a CBIS

- The trend in the computer industry is to produce **smaller**, **faster**, and more **mobile hardware**, such as **tablet** computers.



Components of a CBIS

- **2) Software**:
- consists of the computer programs that govern the operation of the computer.
- software are needed for all types of computers, from small handheld computers to large supercomputers.
- There are **two** primary types of software.
- 1. **System software**, such as Microsoft Windows
- 2. **Application software**, such as Microsoft Excel, allows you to accomplish specific tasks.

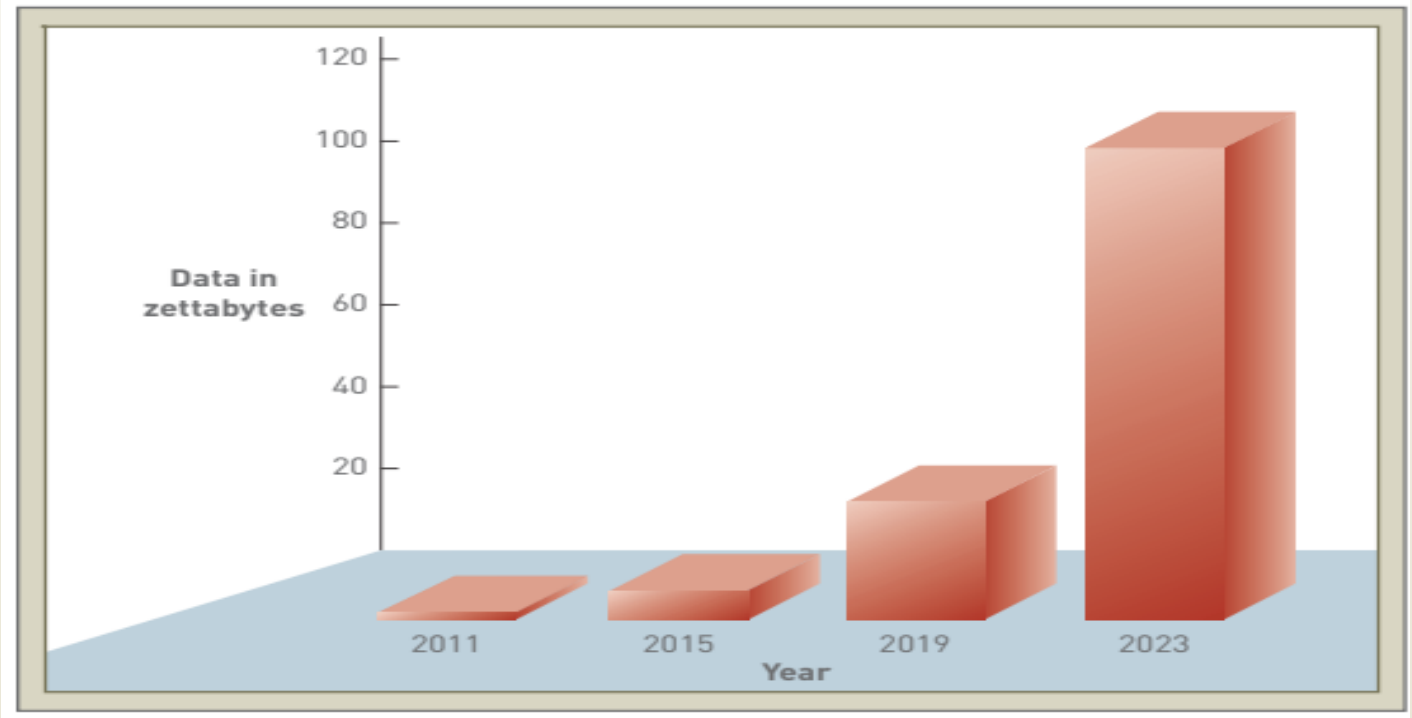
Components of a CBIS

- **3- Databases:** is an organized collection of **facts** and **information**, typically consisting of two or more related data files.
- Scientists estimated that in 2011 there was 1.8 zettabytes (**10^{21}**) of digital data in the world and that the amount of data was doubling every two years.

Components of a CBIS

- The increase in digital data means a huge increase in database storage needs, which will require more storage devices, more space to house the additional storage devices, and additional electricity to operate them.

Growth in data



Components of a CBIS

- The big question is, how will organizations meet the increased demand for data storage?
- Another important issue for any organization is how to keep a vast database secure and safe from the prying eyes of outside individuals and groups.

Components of a CBIS

- **3) Telecommunications, Network and Internet**
- **Telecommunications** is the electronic transmission of signals for communications, which enables organizations to carry out their processes and tasks through effective computer networks.
- Telecommunications can take place through **wired**, **wireless**, and **satellite transmissions**.

Components of a CBIS

- Today, telecommunications is used by people and organizations of all sizes around the world.
- With telecommunications, people can work at home or while traveling. This approach to work, often called “**telework**” or “**telecommuting**,”.
- Telecommunications also enables the use of virtual teams of people working on a project to meet and communicate without ever physically being in the same place.

Components of a CBIS

- **Networks** connect computers and equipment in a building, around the country, or around the world to enable electronic communication.
- Wireless transmission networks enable the use of mobile devices such as smartphones and portable computers.

Components of a CBIS

- The **Internet** is the world's largest computer network, consisting of thousands of interconnected networks, all freely exchanging information.

Components of a CBIS

- A **cloud computing** environment in which software and data storage are provided by the Internet (“**the cloud**”), the services are run on another organization’s computer hardware and both software and data are easily accessed.



Components of a CBIS

- This increased use of the Internet is not without its risks.
- Some people fear that increased usage can lead to problems, including loss of privacy and security, with criminals hacking into an organization's data via the Internet to gain access to sensitive company and customer information.

Components of a CBIS

- The **World Wide Web** (**WWW)**, or the Web, is a network of links on the Internet to documents containing text, graphics, video, and sound.
- Information about the documents and access to them are controlled and provided by tens of thousands of special computers called **Web servers**.
- The **Web** is one of many services available over the Internet and provides access to millions of documents.

Components of a CBIS

- **Intranet:** An internal network based on Web technologies that allows people within an organization to exchange information and work on projects.
- An **extranet** is a network based on Web technologies that allows selected outsiders, such as business partners and customers, to access authorized resources of a company's **intranet**.

Components of a CBIS

- **4- People** are the most important element in computer-based information systems.
- Good systems can enable ordinary people to produce extraordinary results.
- The **chief information officer** (**CIO**), who manages the IS department.

Components of a CBIS

- **5- procedures:** The strategies, policies, methods, and rules for using a **CBIS**.
- Or a **procedure** defines the steps to follow in order to achieve a specific end result.
- Good procedures describe how to achieve the desired end result; who does what and when; and what to do in the event that something goes wrong.

Components of a CBIS

- When people are well trained and follow effective procedures, they can get work done faster, cut costs, make better use of people resources, and enable people to adapt to change.
- When procedures are well documented, they can greatly reduce training costs and shorten the learning curve.

Thanks !

Introduction to Information technology

Lecture 3

Outlines

- Technology
  - Hardware and Software
    - Introduction
    - Hardware Components
    - Processing And Memory Devices: Power, Speed, And Capacity
    - Secondary Storage And Input And Output Devices
    - Input/Output Devices
    - Computer System Types

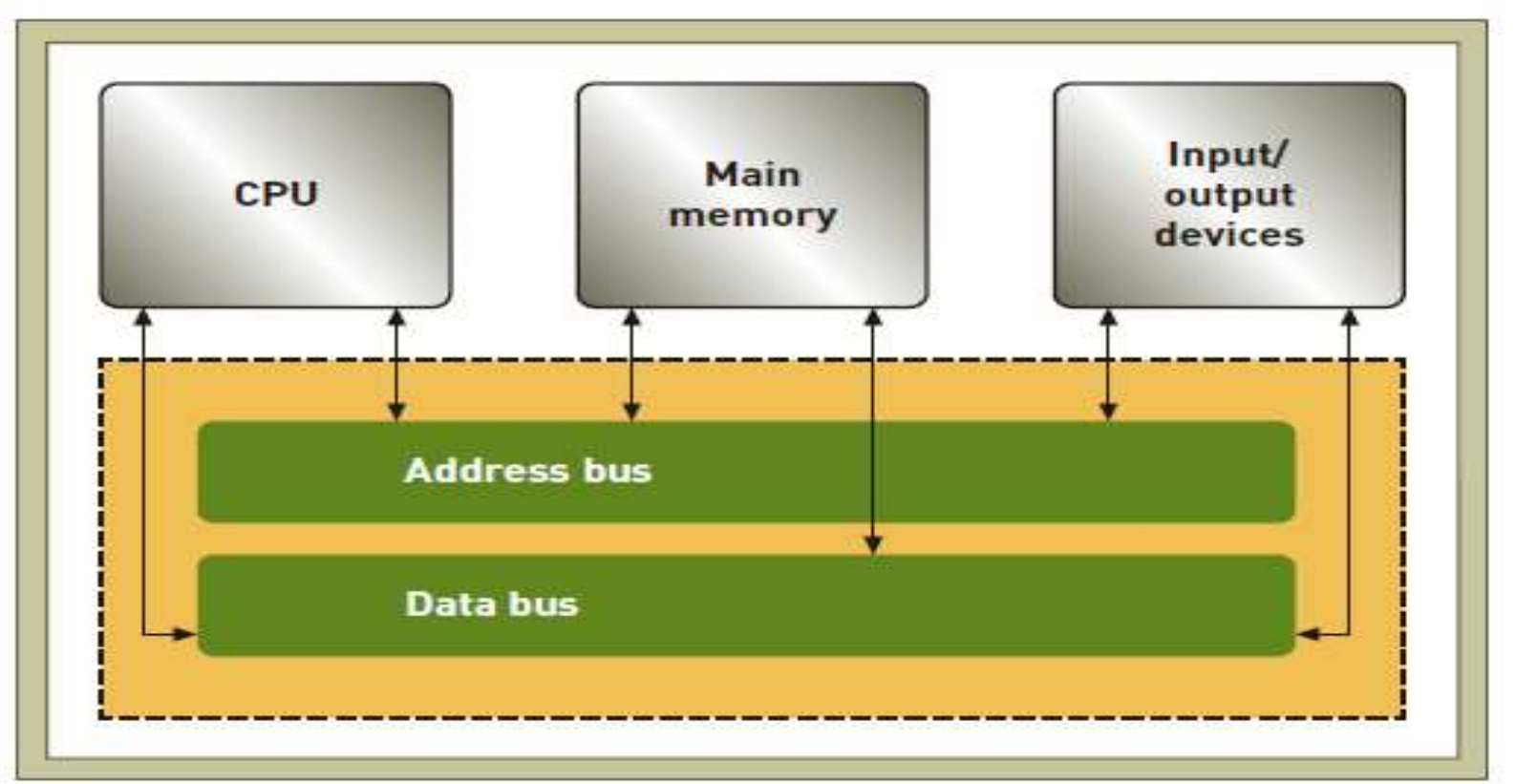
1. Introduction

- Organizations invest in computer hardware and software to improve worker productivity, increase revenue, reduce costs, and provide better customer service.
- choosing the right computer hardware requires understanding its relationship to the information systems and the current and future needs of the organization.

1. Introduction

- **Hardware** refers to the physical components of a computer that perform the input, processing, storage, and output activities of the computer.
- **Software** consists of the computer programs that govern the operation of the computer.

2. HARDWARE COMPONENTS



2. HARDWARE COMPONENTS

- The **central processing unit** (**CPU**) or simply processor is the part of a computer that sequences and executes instructions.
- Central processing unit (**CPU**) consists of three primary elements: the **arithmetic/logic unit**, the **control unit**, and **storage registers**.
- The **arithmetic/ logic unit** (**ALU**) performs mathematical calculations and makes logical comparisons.

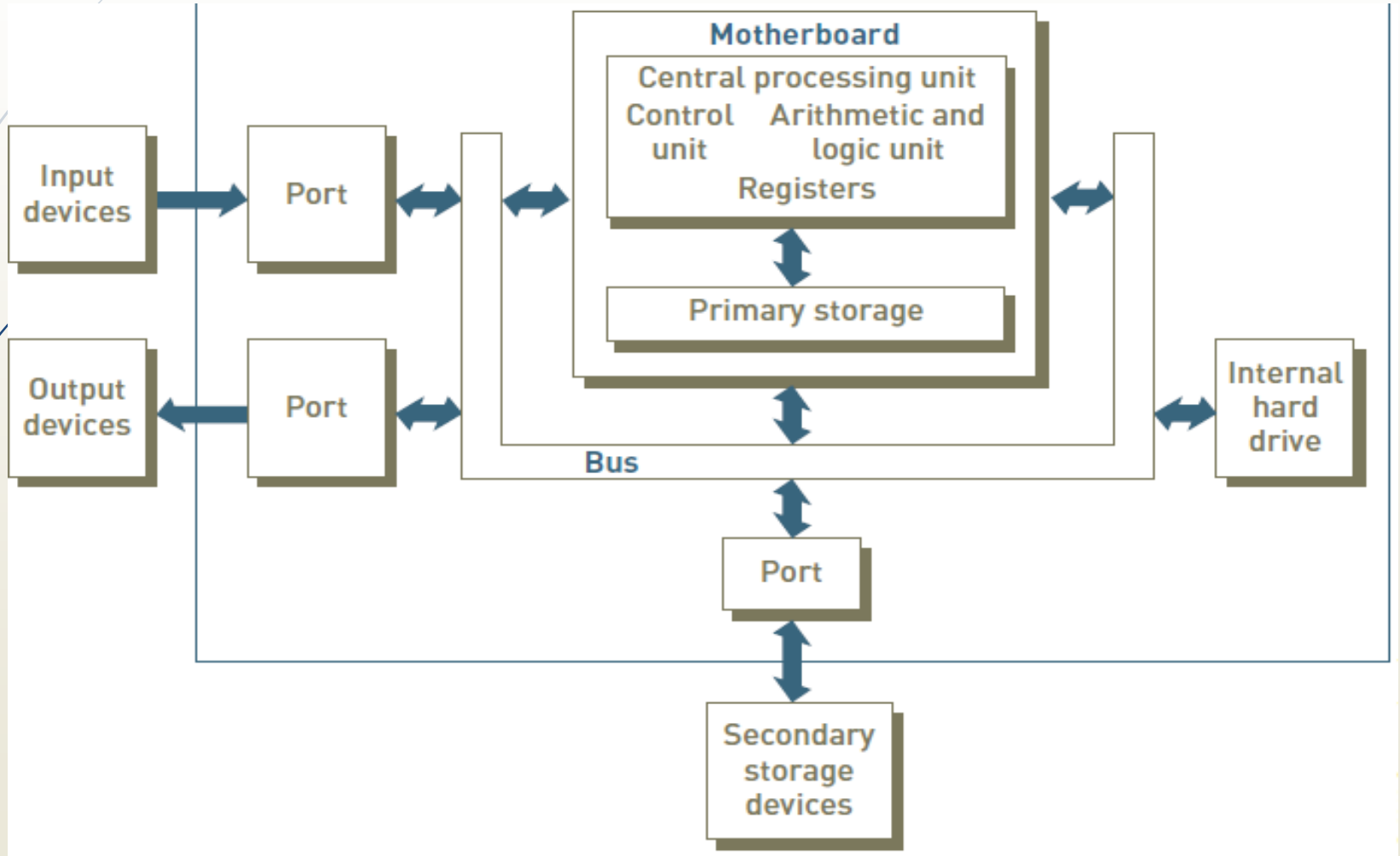
2. HARDWARE COMPONENTS

- The **control unit** sequentially accesses program instructions, decodes them, and coordinates the flow of data in and out of the **ALU, primary storage**, and even **secondary storage** and various output devices.
- **Registers** are high-speed storage areas used to temporarily hold small units of program instructions and data immediately before, during, and after execution by the CPU.

2. HARDWARE COMPONENTS

- **Bus** a set of electronic circuits used to route data and instructions to and from the various components of a computer.
- **Primary storage**, also called **main memory** or **memory**, holds program instructions and data, is closely associated with the CPU.

2. HARDWARE COMPONENTS



3. PROCESSING AND MEMORY DEVICES: POWER, SPEED, AND CAPACITY

- **Processing Characteristics and Functions:**
  - Because efficient processing and timely output are important, organizations use a variety of measures to gauge processing speed.
  - **Clock speed**:
  - A series of electronic pulses produced at a predetermined rate that affects machine cycle time.
  - **Gigahertz** (**GHz**): Billions of cycles per second, a measure of clock speed.

3. PROCESSING AND MEMORY DEVICES: POWER, SPEED, AND CAPACITY

- **Memory Characteristics and Functions :**
- **Memory** provides the CPU with a working storage area for program instructions and data. The chief feature of memory is that it rapidly provides the data and instructions to the CPU.
  - **Storage Capacity**:
  - Like the CPU, memory devices contain thousands of circuits imprinted on a silicon chip.
  - Data is stored in memory as a combination of **on** or **off** circuit states.

3. PROCESSING AND MEMORY DEVICES: POWER, SPEED, AND CAPACITY

- **Memory Characteristics and Functions :**
  - In most cases, storage capacity is measured in bytes, with **1 byte** equivalent to **one character** of data.

3. PROCESSING AND MEMORY DEVICES: POWER, SPEED, AND CAPACITY

- **Memory Characteristics and Functions :**
  - **Types of Memory:**
  - **Random access memory** (RAM):A form of memory in which instructions or data can be temporarily stored.
  - **Read-only memory** (ROM), another type of memory, is **nonvolatile**. In ROM, the combination of circuit states is fixed, and therefore its contents are not lost if the power is removed. ROM provides permanent storage for data and instructions that do not change.

3. PROCESSING AND MEMORY DEVICES: POWER, SPEED, AND CAPACITY

- **Multiprocessing:**
- There are a number of forms of multiprocessing, which involves the simultaneous execution of two or more instructions.
- **Multicore Microprocessors:**
- A microprocessor that combines two or more independent processors into a single computer so that they share the workload and improve processing capacity.

3. PROCESSING AND MEMORY DEVICES: POWER, SPEED, AND CAPACITY

- When selecting a CPU, organizations must balance the benefits of processing speed with energy requirements, size, and cost.
- Chip designers and manufacturers are exploring various means to avoid heat problems in their new designs.
- **ARM** is a computer chip design company whose energy-efficient chip architecture is broadly used in smartphones and tablet computers.

3. PROCESSING AND MEMORY DEVICES: POWER, SPEED, AND CAPACITY

- **Multiprocessing:**
  - **Parallel Computing:**
  - The simultaneous execution of the same task on multiple processors to obtain results faster.
  - The most frequent uses for parallel computing include modeling, simulation, and analyzing large amounts of data.

3. PROCESSING AND MEMORY DEVICES: POWER, SPEED, AND CAPACITY

- **Multiprocessing:**
- **Massively parallel processing system**: A form of multiprocessing that speeds processing by linking hundreds or thousands of processors to operate at the same time
- **Grid Computing:**
  - The use of a collection of computers, often owned by multiple individuals or organizations, to work in a coordinated manner to solve a common problem.
  - Grid computing is low-cost approach to parallel processing.

3. PROCESSING AND MEMORY DEVICES: POWER, SPEED, AND CAPACITY

- **Multiprocessing:**
- **Grid Computing:**
  - Key to the success of grid computing is a central server that acts as the grid leader and traffic monitor.

4. SECONDARY STORAGE AND INPUT AND OUTPUT DEVICES

- Computer systems also need to store larger amounts of data, instructions, and information more permanently than main memory allows.
- Secondary storage, also called permanent storage, serves this purpose.

4. SECONDARY STORAGE AND INPUT AND OUTPUT DEVICES

- **Access Methods:**
  - Data and information access can be either **sequential** or **direct**.
  - **Sequential access:** means that data must be accessed in the order in which it is stored.
  - **Direct access:** means that data can be retrieved directly, without having to pass by other data in sequence.
  - **Sequential access storage device (SASD):** A device used to sequentially access secondary storage data
  - **Direct access storage device (DASD):** A device used for direct access of secondary storage data.

4. SECONDARY STORAGE AND INPUT AND OUTPUT DEVICES

- **Secondary Storage Devices**
- **Magnetic Tapes:**
- A type of sequential secondary storage medium, now used primarily for storing backups of critical organizational data in the event of a disaster
- **Hard Disks:**
- A **hard disk drive** (**HDD**) is a direct access storage device used to store and retrieve data from rapidly rotating disks coated with magnetic material.

4. SECONDARY STORAGE AND INPUT AND OUTPUT DEVICES

- Secondary Storage Devices



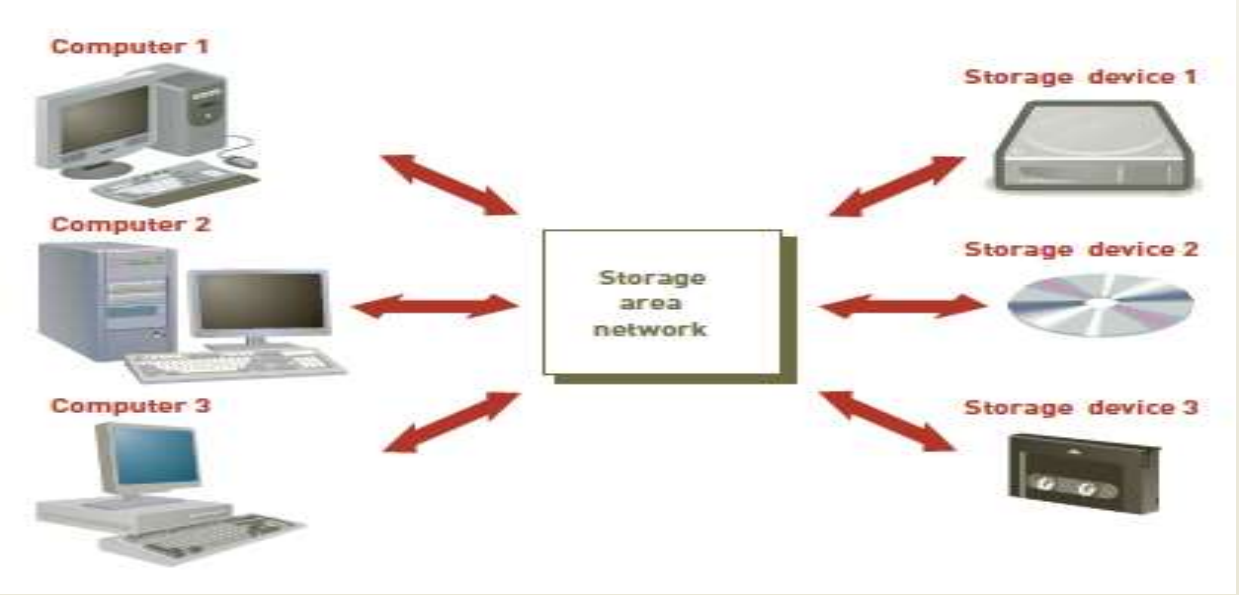


4. SECONDARY STORAGE AND INPUT AND OUTPUT DEVICES

- **RAID:**
- A **r**edundant **a**rray of **i**ndependent/**i**nexpensive **d**isks (RAID) is a method of storing data that generates extra bits of data from existing data, enabling the system to create a "***reconstruction map***" so that if a hard drive fails, it can rebuild lost data.
- **Disk mirroring:** A process of storing data that provides an exact copy that protects users fully in the event of data loss.

4. SECONDARY STORAGE AND INPUT AND OUTPUT DEVICES

- **Storage Area Networks:**
- high-speed, special-purpose network that integrates different types of data storage devices (e.g., HDD, magnetic tape, SSD) into a single storage system and connects that to computing resources across an entire organization.



4. SECONDARY STORAGE AND INPUT AND OUTPUT DEVICES

- **Optical Secondary Storage Devices:**
- An optical storage device uses special lasers to read and write data.
- **Solid State Secondary Storage Devices (SSD):**
- A storage device that stores data in memory chips rather than on hard disk drives or optical media.
- These memory chips require less power and provide much faster data access than magnetic data storage devices.

4. SECONDARY STORAGE AND INPUT AND OUTPUT DEVICES

- **Storage as a Service:**
- A data storage model where a data storage service provider rents space to individuals and organizations.
- Users access their rented data storage via the Internet.

5. Input/output devices

- **Input/output devices:**
- provide data and instructions to the computer and receive results from it.

6. COMPUTER SYSTEM TYPES

- **Portable Computers:**
- A computer small enough to carry easily.
- **Laptops**:
  - A personal computer designed for use by mobile users, being small and light enough to sit comfortably on a user's lap.
- **Tablet computer:**
  - A portable, lightweight computer with no keyboard that allows you to roam the office, home, or factory floor carrying the device like a clipboard.

6. COMPUTER SYSTEM TYPES

- **Nonportable Single-User Computers:**
- **Thin client:**
  - is a low-cost, centrally managed computer with no internal or external attached drives for data storage.
  - Thin clients work well in a cloud-computing environment to enable users to access the computing and data resources available within the cloud.

6. COMPUTER SYSTEM TYPES

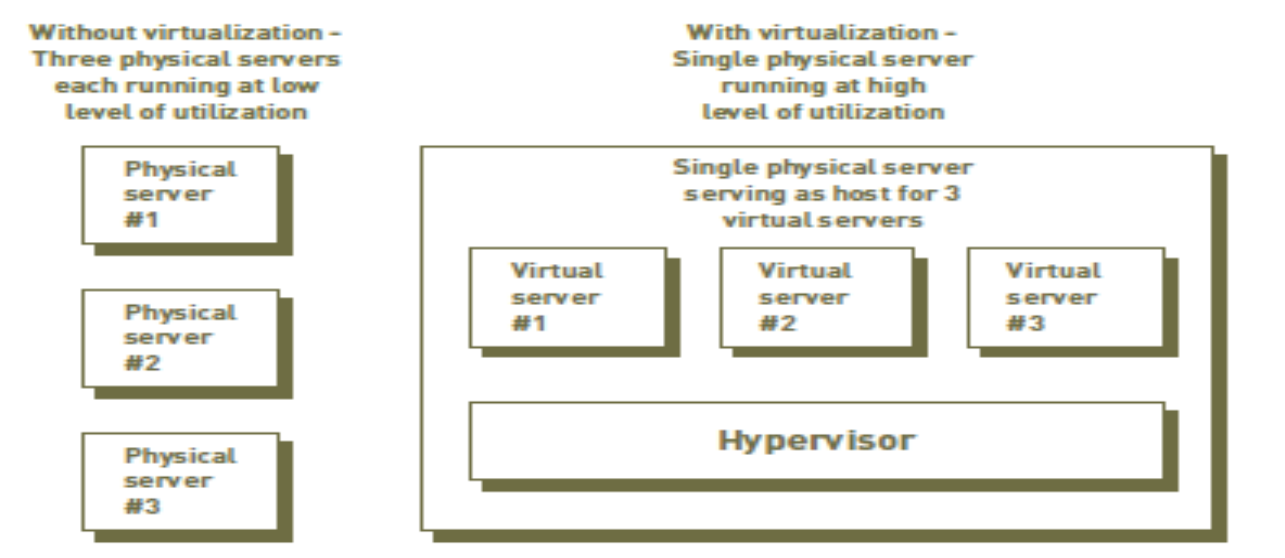
- **Nonportable Single-User Computers:**
  - **Desktop computers:**
    - A non-portable computer that fits on a desktop and provides sufficient computing power, memory, and storage for most business computing tasks.
  - **Workstations:**
    - A more powerful personal computer used for mathematical computing, computer-assisted design, and other high-end processing but still small enough to fit on a desktop.

6. COMPUTER SYSTEM TYPES

- **Multiple-User Computer Systems:**
  - **Server:**
    - A **server** is a computer employed by many users to perform a specific task, such as running network or Internet applications.
  - **Server farm:**
    - A facility that houses a large number of servers in the same room, where access to the machines can be controlled and authorized support personnel can more easily manage and maintain the servers.

6. COMPUTER SYSTEM TYPES

- **Multiple-User Computer Systems:**
  - **virtual server:**
    - A method of logically dividing the resources of a single physical server to create multiple logical servers, each acting as its own dedicated machine.



6. COMPUTER SYSTEM TYPES

- **Multiple-User Computer Systems:**
  - **Mainframe computer:**
    - A **mainframe computer** is a large, powerful computer shared by dozens or even hundreds of concurrent users connected to the machine over a network.
  - **Supercomputers:**
    - **Supercomputers** are the most powerful computers with the fastest processing speed and highest performance.

6. COMPUTER SYSTEM TYPES

Multiple-User Computer Systems:

The supercomputer Tianhe-2 can operate as fast as 33.86 petaflops per second, and is currently ranked as the world's fastest computer.



6. COMPUTER SYSTEM TYPES

- **DATA CENTERS:**
  - A **data center** is a climate-and-access-controlled building or a set of buildings that houses the computer hardware that delivers an organization’s data and information services.

6. COMPUTER SYSTEM TYPES

GREEN COMPUTING:

- **Green computing**: A program concerned with the efficient and environmentally responsible design, manufacture, operation, and disposal of IS-related products.

Thanks !

Introduction to Information Technology

Lecture 4

Outlines

- 1. An OVERVIEW OF SOFTWARE
  - 1.1 Systems Software
    - Operating system
    - Utililty Software
    - Middleware Software
  - 1.2 Application Software
    - Types of APPLICATION SOFTWARE

1. An OVERVIEW OF SOFTWARE

- **Software** consists of computer programs that control the workings of computer hardware.
- Software can be divided into two types: **systems software** and **application software**.

1. An OVERVIEW OF SOFTWARE

- **System software** includes ***operating systems***, ***utilities***, and ***middleware*** that coordinate the activities and functions of the hardware and other programs throughout the computer system.
- **Application software** consists of programs that help users solve particular computing problems.

1.1 Systems Software

- The primary role of system software is to control the operations of computer hardware.
- System software also supports the problem-solving capabilities of application programs.

1.1 Systems Software

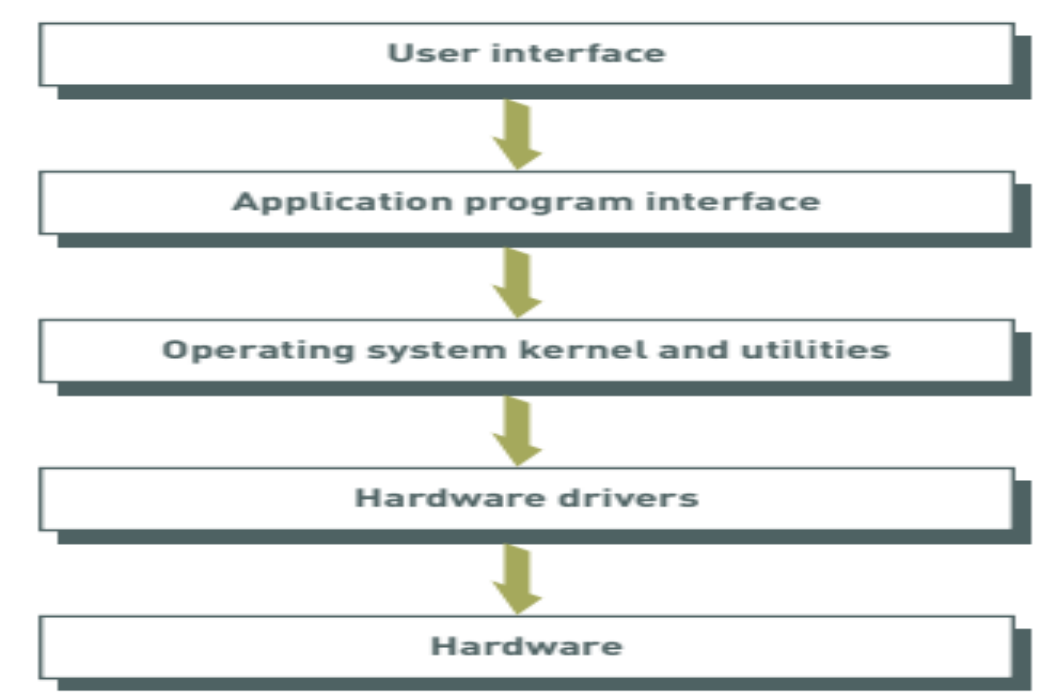
- **Operating Systems**:
  - A set of computer programs that controls the computer hardware and acts as an interface to application software.

1.1 Systems Software

- **Operating Systems:**
- The various combinations of OSs, computers, and users include the following:
  - 1- Single computer with a single user.
  - 2- Single computer with multiple simultaneous users. (Server or mainframe computers, e.g **UNIX**)
  - 3- Multiple computers with multiple users. (Computer networks, including home networks e.g **Network server OSs**)
  - 4- Special-purpose computers. (control sophisticated military aircraft, digital cameras)

1.1 Systems Software

- **Operating Systems**:
  - **kernel**: The heart of the operating system that controls the most critical processes of the OS.

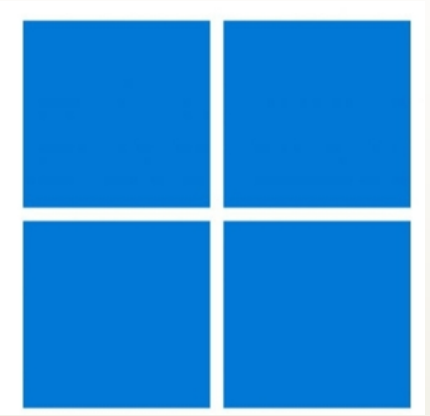


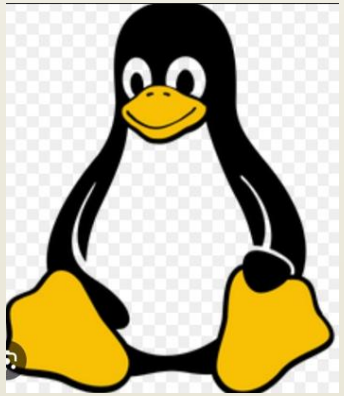
1.1 Systems Software

- **Operating Systems**:
  - **Personal Computing Operating Systems**
  - An operating systems that are found on personal computers, portable computers, and mobile devices.

1.1 Systems Software

- **Operating Systems:**
  - **Some Common Operating Systems**
    - **Microsoft Windows**
    - **Mac OS**
    - **Linux**









1.1 Systems Software

- **Operating Systems:**
  - **Workgroup Operating Systems**
  - Powerful and sophisticated operating systems are needed to run the servers that meet these business needs for workgroups.
    - ✓ **UNIX**
    - ✓ **Windows Server**
    - ✓ **Red Hat Linux**
    - ✓ **Mac OS X Server**

1.1 Systems Software

- **Utility Programs:**
  - A program that helps to perform maintenance or correct problems with a computer system.

1.1 Systems Software

- **Middleware:**
  - is software that provides messaging services that allow different applications to communicate and exchange data.
  - This systematic tying together of disparate applications, often through the use of middleware, is known as **enterprise application integration** (**EAI**).
  - **Service-oriented architecture** (**SOA**) is a software design approach based on the use of discrete pieces of software (modules) to provide specific functions as services to other applications.

1.2 APPLICATION SOFTWARE

- The primary function of application software is to apply the power of a computer system to enable people, workgroups, and entire organizations to solve problems and perform specific tasks.
- **Proprietary software** and **off-the-shelf software** are two important types of application software.

Types of APPLICATION SOFTWARE

- **Proprietary software** is one-of-a-kind software designed for a specific application and owned by the company, organization, or person that uses it.
- **Off-the-shelf software** is produced by software vendors to address needs that are common across businesses, organizations, or individuals.

Types of APPLICATION SOFTWARE

- Organizations typically use **off-the-shelf application** software to meet common business needs and **proprietary application software** to meet unique business needs and provide a competitive advantage.

Types of APPLICATION SOFTWARE

- **Software as a service** (**SaaS**): A service that allows businesses to subscribe to Web-delivered application software.
- **Personal Application Software**: software, includes the general-purpose tools and programs that support individual needs.

Types of APPLICATION SOFTWARE

- **Workgroup Application Software:** Software that supports teamwork, whether team members are in the same location or dispersed around the world.
- **Enterprise Application Software**: Software that benefits an entire organization enterprise application software can be developed specifically for the business or purchased off the shelf.
- There are many categories of enterprise software, including the following:

Types of APPLICATION SOFTWARE

Accounts payable	Invoicing
Accounts receivable	Manufacturing control
Airline industry operations	Order entry
Automatic teller systems	Payroll
Cash-flow analysis	Receiving
Check processing	Restaurant management
Credit and charge card administration	Retail operations
Distribution control	Sales ordering
Fixed asset accounting	Savings and time deposits
General ledger	Shipping
Human resource management	Stock and bond management
Inventory control	Tax planning and preparation

Types of APPLICATION SOFTWARE

- **Programming Languages**: Sets of keywords, commands, symbols, and rules for constructing statements by which humans can communicate instructions to a computer.
- **syntax**: A set of rules associated with a programming language.

Thanks !

Introduction to I.T

Lecture 5

Topics:

- 1. Introduction to Database
  - 1.1 Database
  - 1.2 Examples of Database
  - 1.3 DBMS
- 2. Data Fundamentals
  - 2.1 Hierarchy of Data
  - 2.2 Data Entities, Attributes, and Keys
  - 2.3 The Database Approach

1. Introduction to Database

1.1 Database

- A **database** is a well-designed, organized, and carefully managed collection of data.
- **Another defintion**: Collection of related data.

1.2 Examples of Database

- **Traditional database applications**
  - Store textual or numeric information.
- **Multimedia databases**
  - Store images, audio clips, and video streams digitally.
- **Geographic information systems (GIS)**
  - Store and analyze maps, weather data, and satellite images.

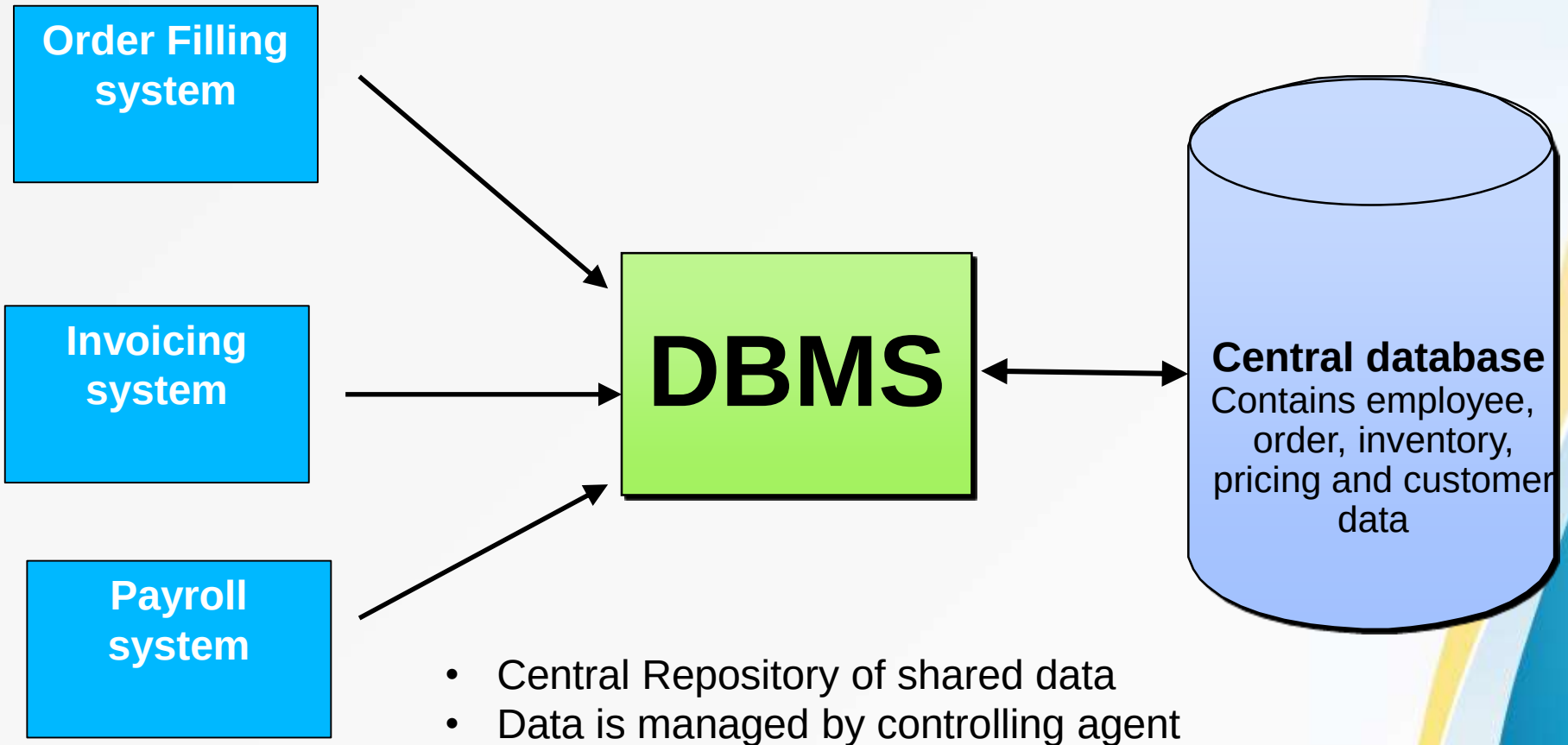
1.2 Examples of Database

- **Data warehouses and online analytical processing (OLAP) systems**
  - Extract and analyze useful business information from very large databases.
  - Support decision making.
- **Real-time and active database technology**
  - Control industrial and manufacturing processes.

1.3 DBMS

- **database management system** (**DBMS**): A group of programs used to access and manage a database as well as provide an interface between the database and its users and other application programs.
- Another **definition** The **DBMS** is a *general-purpose software system* that facilitates the processes of ***defining***, ***constructing***, ***manipulating***, and ***sharing*** databases among various users and applications.

1.3 DBMS



- Central Repository of shared data
- Data is managed by controlling agent
- Stored in a standardized, convenient form

2. Data Fundamentals

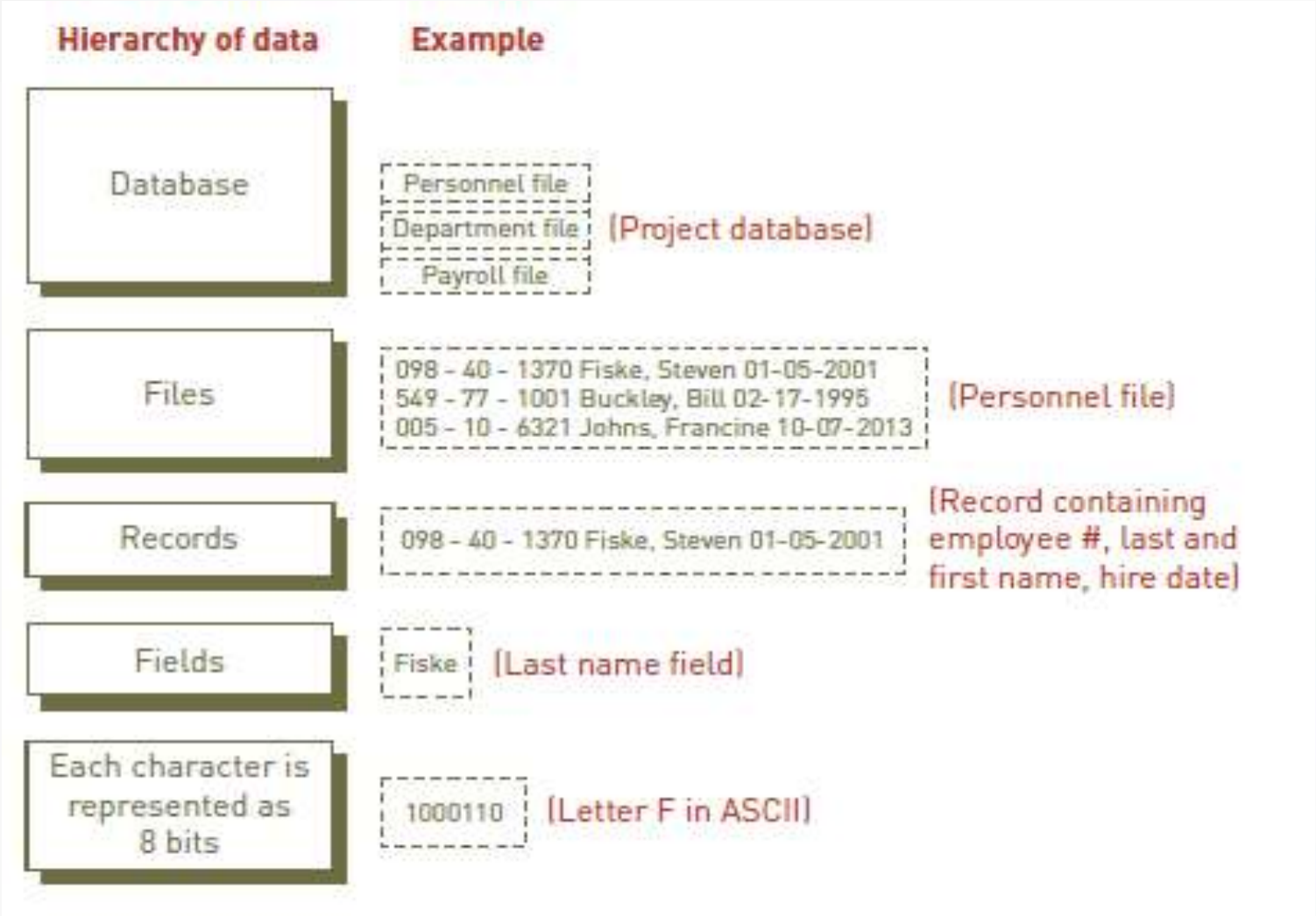
2.1 Hierarchy of Data

- **bit**: A binary digit (i.e., **0** or **1**) that represents a circuit that is either **on** or **off**.
- **character**: A basic building block of most information, consisting of *uppercase letters*, *lowercase letters*, *numeric digits*, or *special symbols*.
- **field**: Typically, a name, a number, or a combination of characters that describes an aspect of a business object or activity

2.1 Hierarchy of Data

- **record**: A collection of data fields all related to one object, activity, or individual.
- **file**: A collection of related records.
- At the highest level of the data hierarchy is a **database**, a collection of integrated and related files. Together, bits, characters, fields, records, files, and databases form the **hierarchy of data**.
- **Characters** are combined to make a **field**, **fields** are combined to make a **record**, **records** are combined to make a **file**, and **files** are combined to make a **database**.

2.1 Hierarchy of Data

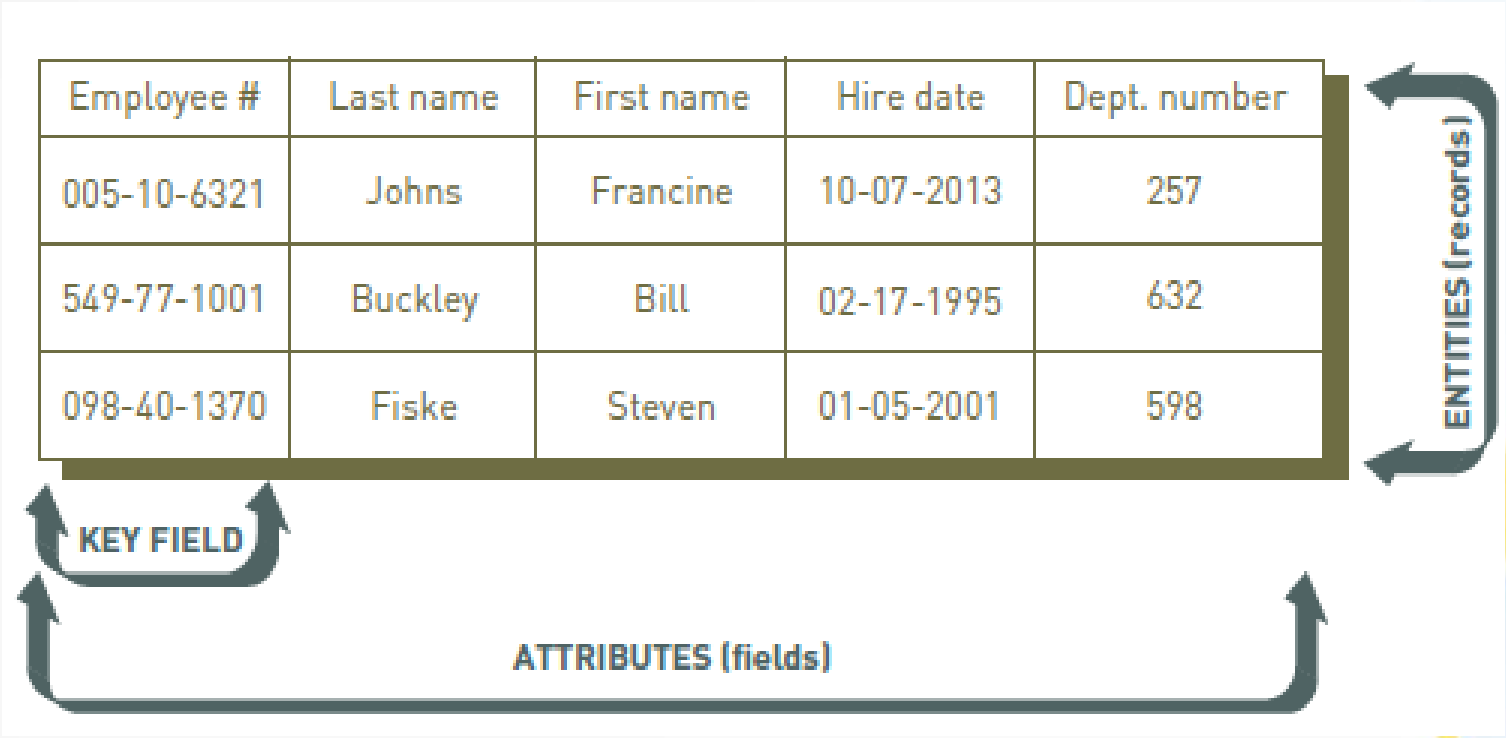


2.2 Data Entities, Attributes, and Keys

- **entity**: A person, place, or thing for which data is collected, stored, and maintained.
- **attribute**: is a characteristic of an entity.
- **data item**: The specific value of an attribute.

2.2 Data Entities, Attributes, and Keys

- The **key** field is the employee number.
- The **attributes** include last name, first name, hire date, and department number.



2.2 Data Entities, Attributes, and Keys

- **primary key**: A field or set of fields that uniquely identifies the record.
- The primary key is used to **distinguish** records so that they can be accessed, organized, and manipulated.

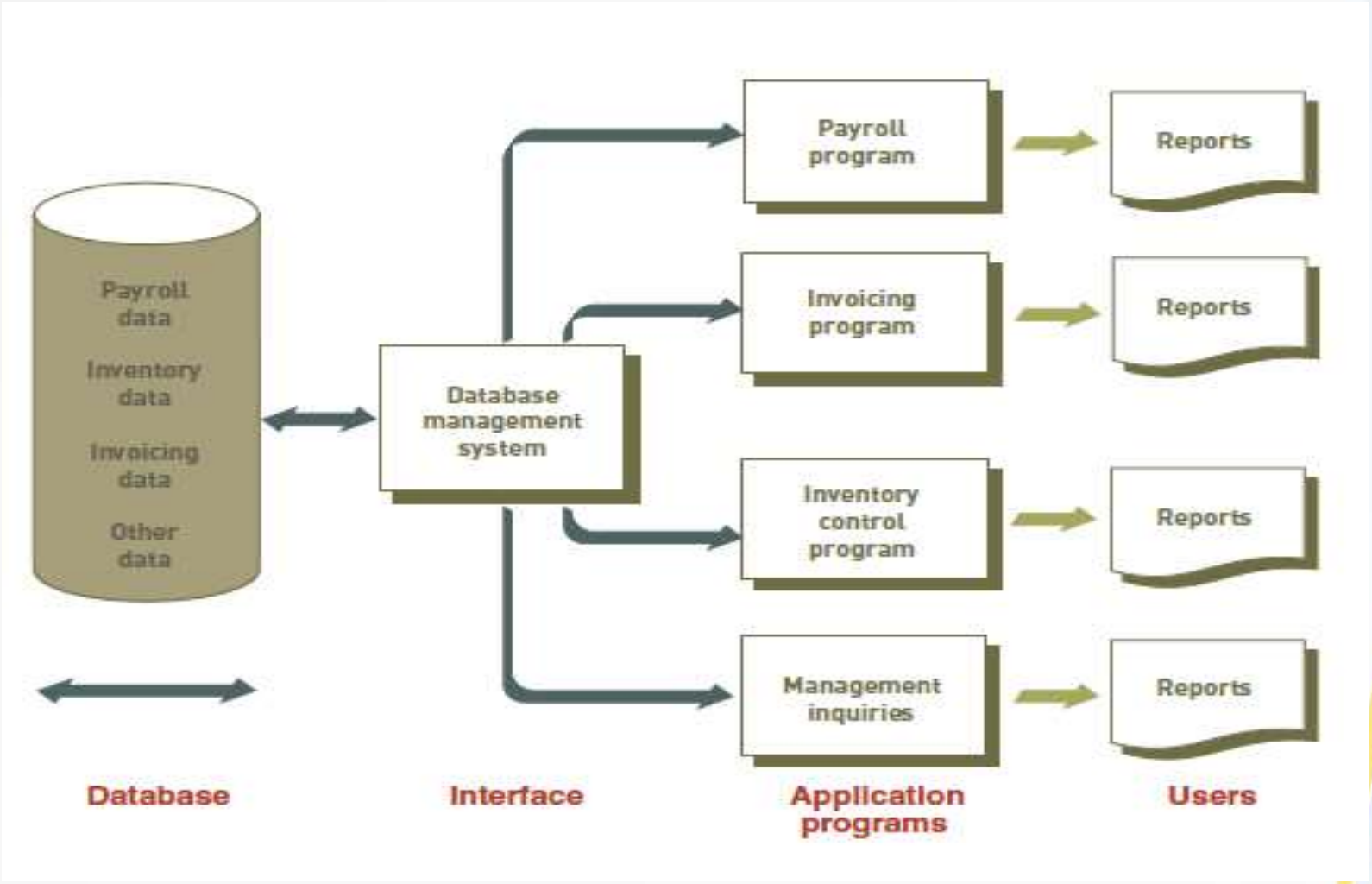
2.3 The Database Approach

- **database approach to data management**: An approach to data management where multiple information systems share a **pool** of related data.
- A database offers the ability to share data and information resources.
- To use the **database approach to data management**, additional software—a database management system (**DBMS**)—is required.

2.3 The Database Approach

- As previously discussed, a **DBMS** consists of a group of programs that can be used as an ***interface*** between a database and the user of the database.
- Typically, this software acts as a **buffer** between the application programs and the database itself.

2.3 The Database Approach



Questions?

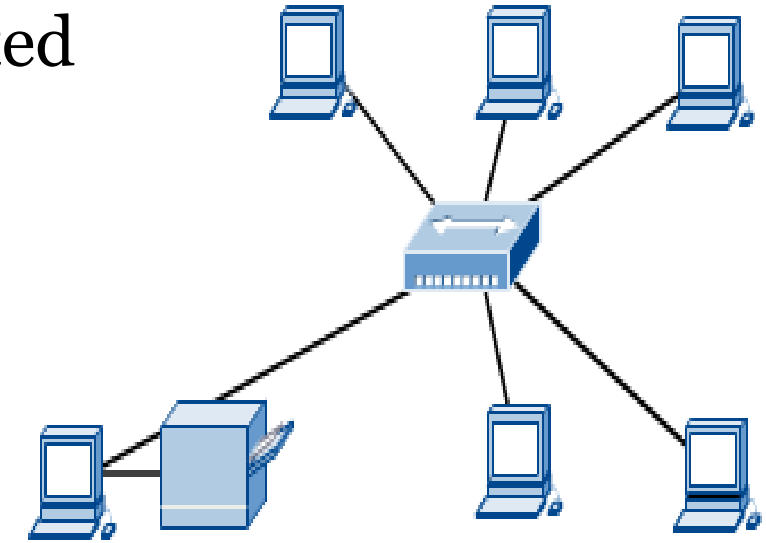
Introduction to I.T

Introduction to Network part 1

Lecture 6

Computer Networks

- A computer network consists of 2 or more computers connected together, and they can communicate and share resources.
- Recourses can be:
  - Hardware, Software ,Data/Information ,... etc.

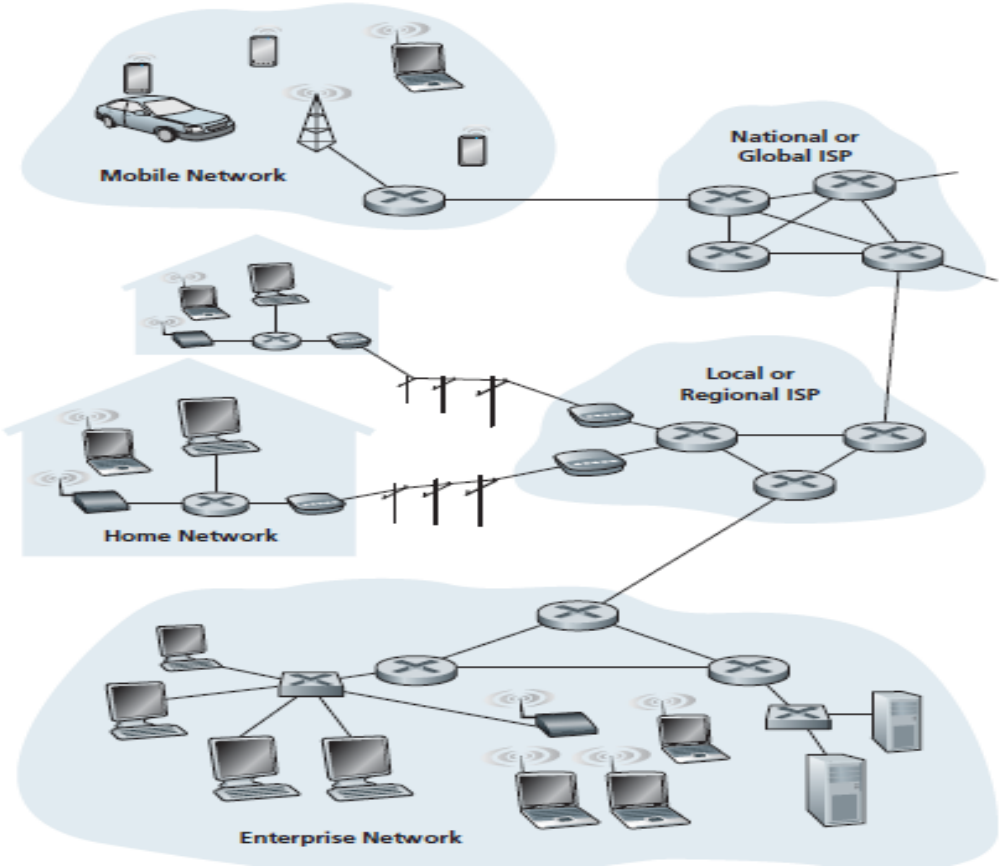


Computer Networks

- A network is a set of devices(often referred to as nodes). Connected by communication links.
  - A node can be “a computer, printer, or any other device capable of sending and/or receiving data generated by other nodes on the network”.
- A network must be able to meet a certain number of criteria:
  - Performance,
  - Reliability,
  - Standardization
  - And security.

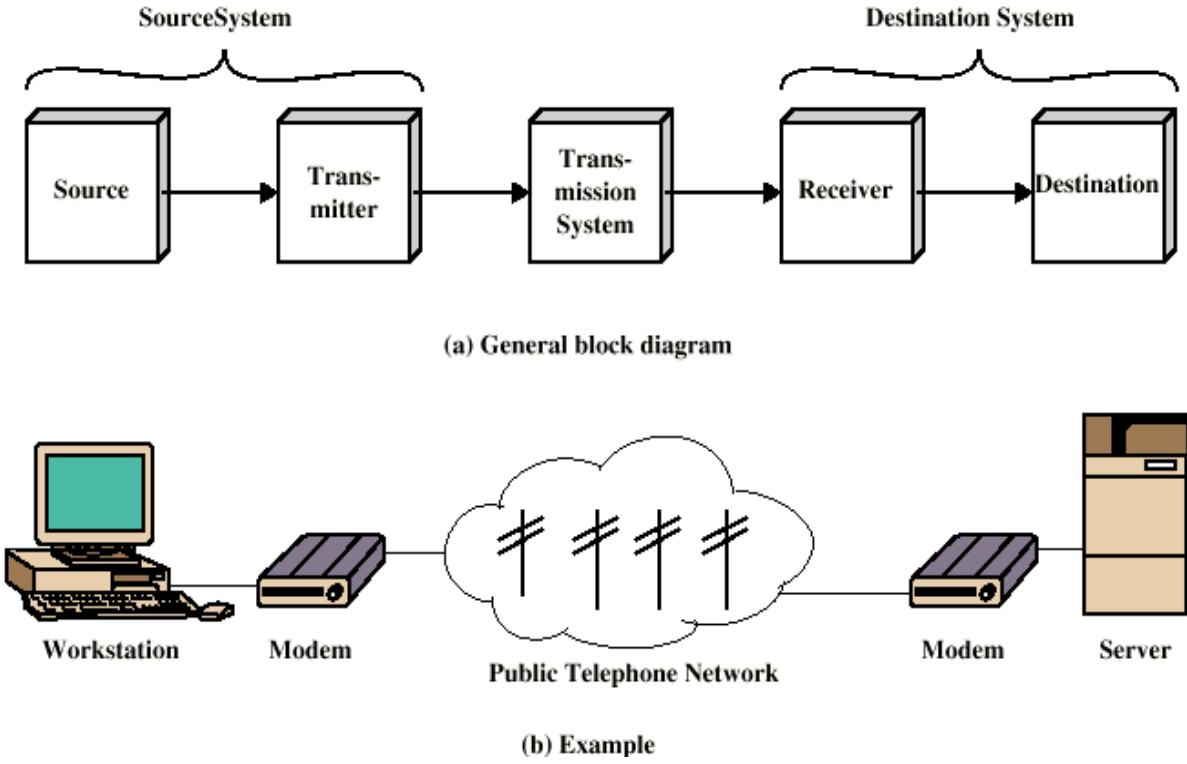
Computer Networks

- Computers may connect to each other by either wired or wireless media.



Computer Networks

- Simplified Communications Model - Diagram



Computer Networks

- Source
  - Generates data to be transmitted
- Transmitter
  - Converts data into transmittable signals
- Transmission System
  - Carries data
- Receiver
  - Converts received signal into data
- Destination
  - Takes incoming data

Uses of Computer Networks

- Business Applications
  - Sharing recourses.
  - E-commerce (electronic commerce).
- Home Applications
  - Access to remote information.
  - Person-to-person communication.
  - Interactive entertainment.
  - Electronic commerce.
- Mobile Users
- Social Issues

Need of Computer Networks

- To provide sharing of resources such as information or processors.
- To provide distribution of processing functions.
- To provide centralized control for a geographically distributed system.
- To provide centralized management and allocation of network resources.

Need of Computer Networks

- To provide compatibility of dissimilar equipment and software.
  - To provide network users with maximum performance at minimum cost.
- To provide an efficient means of transport large volumes of data among remote locations.

Advantages of Computer Networks

- Resource Sharing
- High Reliability
- Low Cost/Saving Money
- Communications
- Internet Access Sharing
- Data Security and Management

Disadvantages of Computer Networks

- Network Hardware, Software and Setup Costs
- Hardware and Software Management and Administration Costs
- Undesirable Sharing
- Illegal or Undesirable Behavior
- Data Security Concerns

Classification of Computer Networks

- Computer networks are classified based on various factors. They include:
  - Geographical span
  - Inter-connectivity
  - Administration
  - Architecture

Geographical Span

- Geographically a network can be seen in one of the following categories:
  - It may be spanned across your table, among Bluetooth enabled devices, Ranging not more than few meters.
  - It may be spanned across a whole building, including intermediate devices to connect all floors.
  - It may be spanned across a whole city.
  - It may be spanned across multiple cities or provinces.
  - It may be one network covering whole world.

Inter-Connectivity

- Components of a network can be connected to each other differently in some fashion. By connectedness we mean either logically, physically, or both ways.
  - Every single device can be connected to every other device on network, making the network mesh.
  - All devices can be connected to a single medium but geographically disconnected, created bus-like structure.
  - Each device is connected to its left and right peers only, creating linear structure.
  - All devices connected together with a single device, creating star-like structure.
  - All devices connected arbitrarily using all previous ways to connect each other, resulting in a hybrid structure.

Administration

- A network can be private network while belongs a single autonomous system and cannot be accessed outside its physical or logical domain.
- A network can be public, while is accessed by all.
- Three forms:
  - Intranet
  - Extranet
  - Internet

Network Architecture

- Computer networks can be discriminated into various types such as Client-Server, peer-to-peer or hybrid, depending upon its architecture.
  - There can be one or more systems acting as Server. Other being Client, requests the Server to serve requests. Server takes and processes request on behalf of Clients.
  - Two systems can be connected Point-to-Point, or in back-to-back fashion. They both reside at the same level and called peers.
  - There can be hybrid network while involves network architecture of both the above types.

Network Applications: examples

- Computer systems and peripherals are connected to form a network. They provide numerous advantages:
  - Resource sharing such as printers and storage devices.
  - Exchange of information by means of e-Mails and File Transfer.
  - Information sharing by using Web or Internet.
  - Interaction with other users using dynamic web pages.
  - IP phones.
  - Video conferences.
  - Parallel computing.
  - Instant messaging.

See you next week

Introduction to Information Technology

Lecture 7

Introduction to Network part 2

- Types of Networks
- Network Topologies
- Network Devices

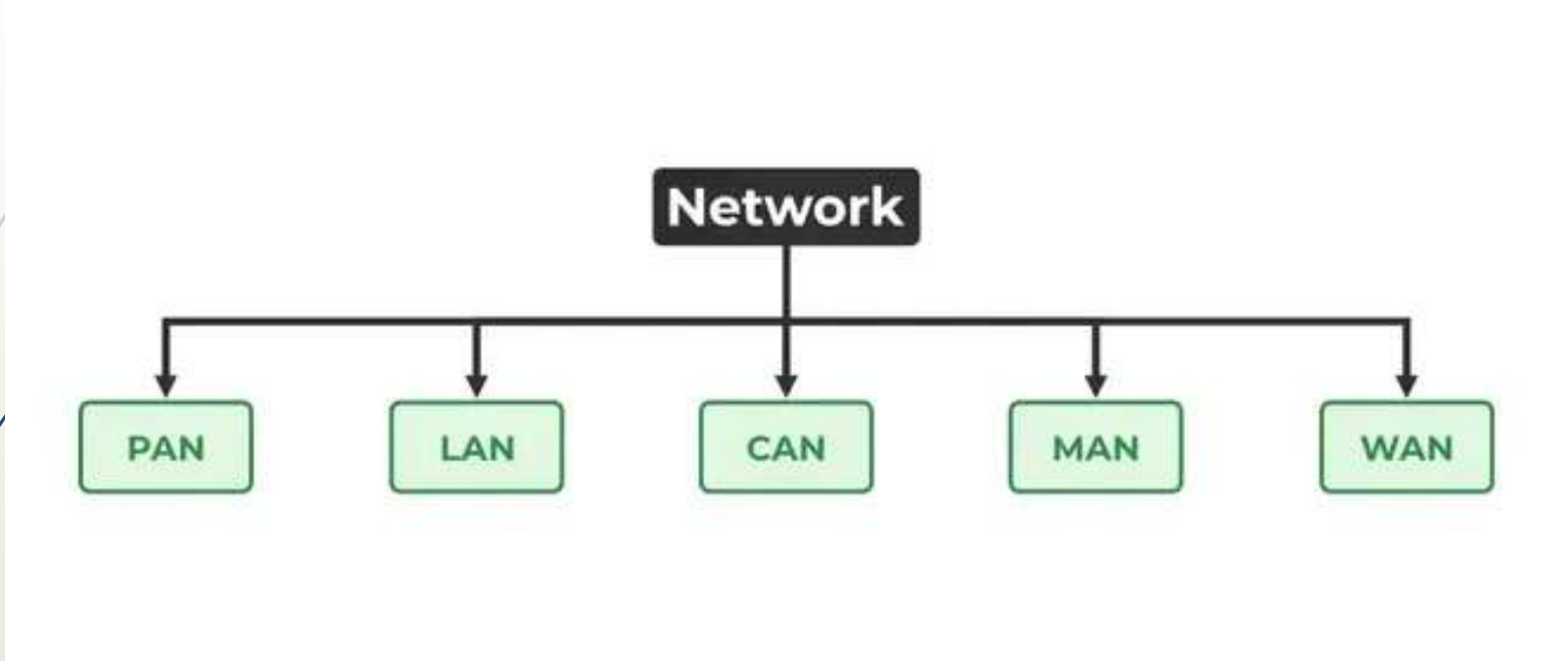
Types of Networks

- Types of Computer Networks are classified based on several factors:

1. **Geographical area**
2. **Transmission medium**
3. **Ownership and access control**

- **Classification Based on Geographical Area**
- Categorizing computer networks according to the physical distance or area they cover, ranging from small personal spaces to large global regions. Based on geographical area, computer networks are mainly divided into five types:

Types of Networks

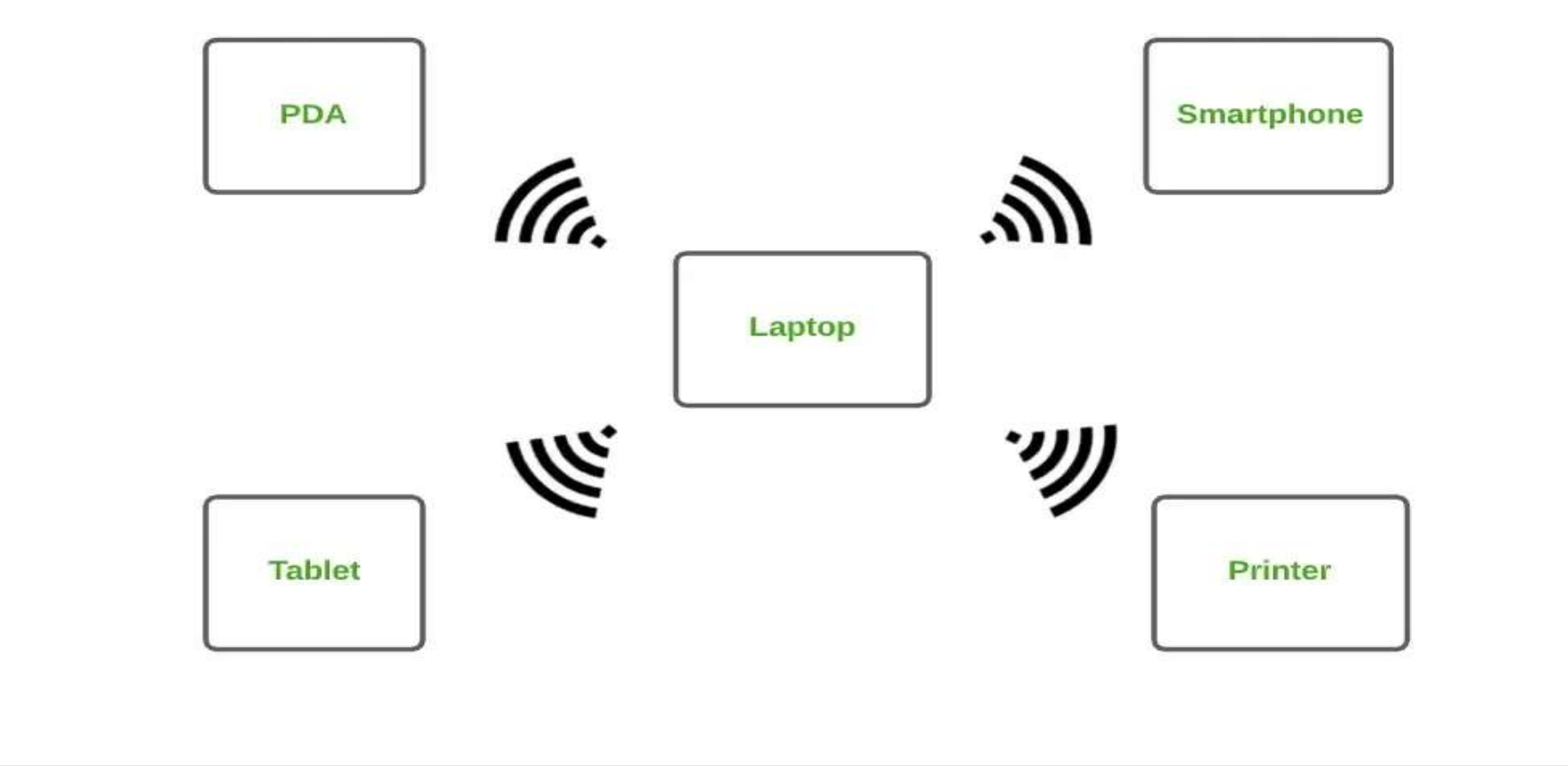


Classification Based on Geographical Area

1. Personal Area Network (PAN)

- This connects personal devices within a very short range around a single user.
- **Coverage Range:** 1–10 meters
- **Key Technologies:** Bluetooth, NFC, Infrared
- **Real-World Example:** Smartphone connected to wireless earbuds
- **When It Is Used:** For short-range communication between personal devices

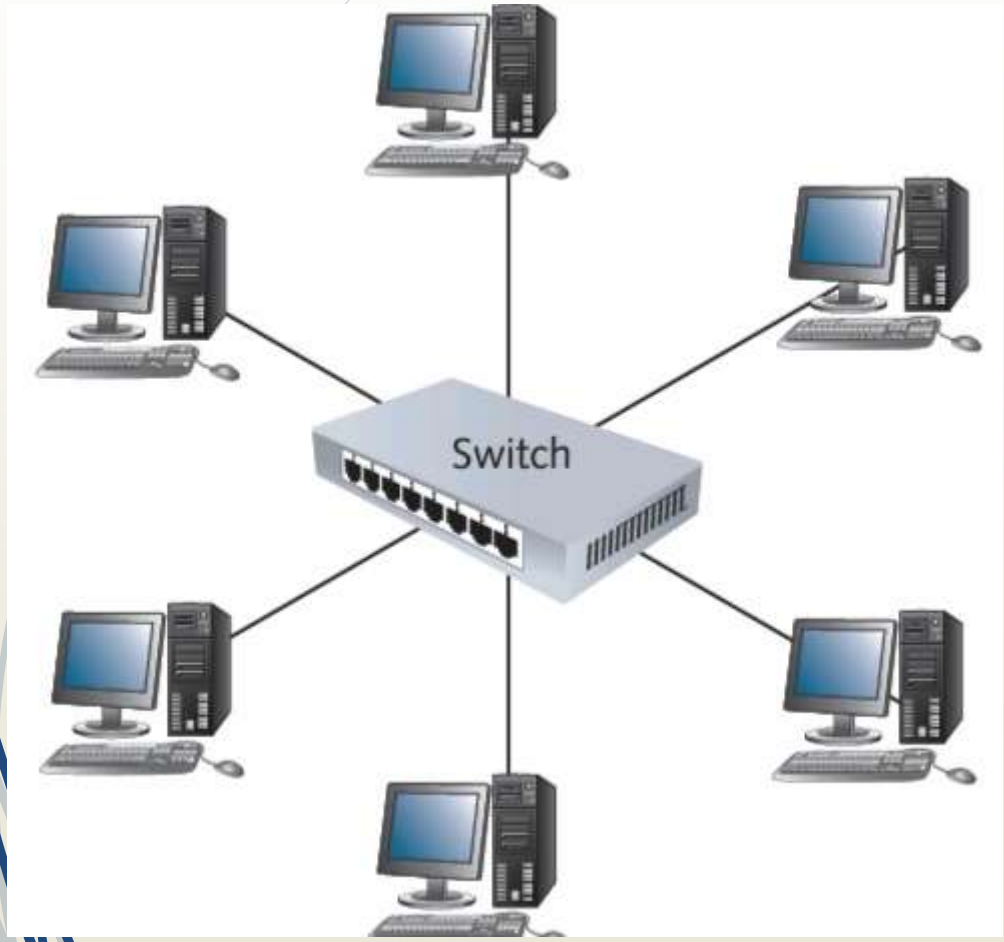
1. Personal Area Network (PAN)

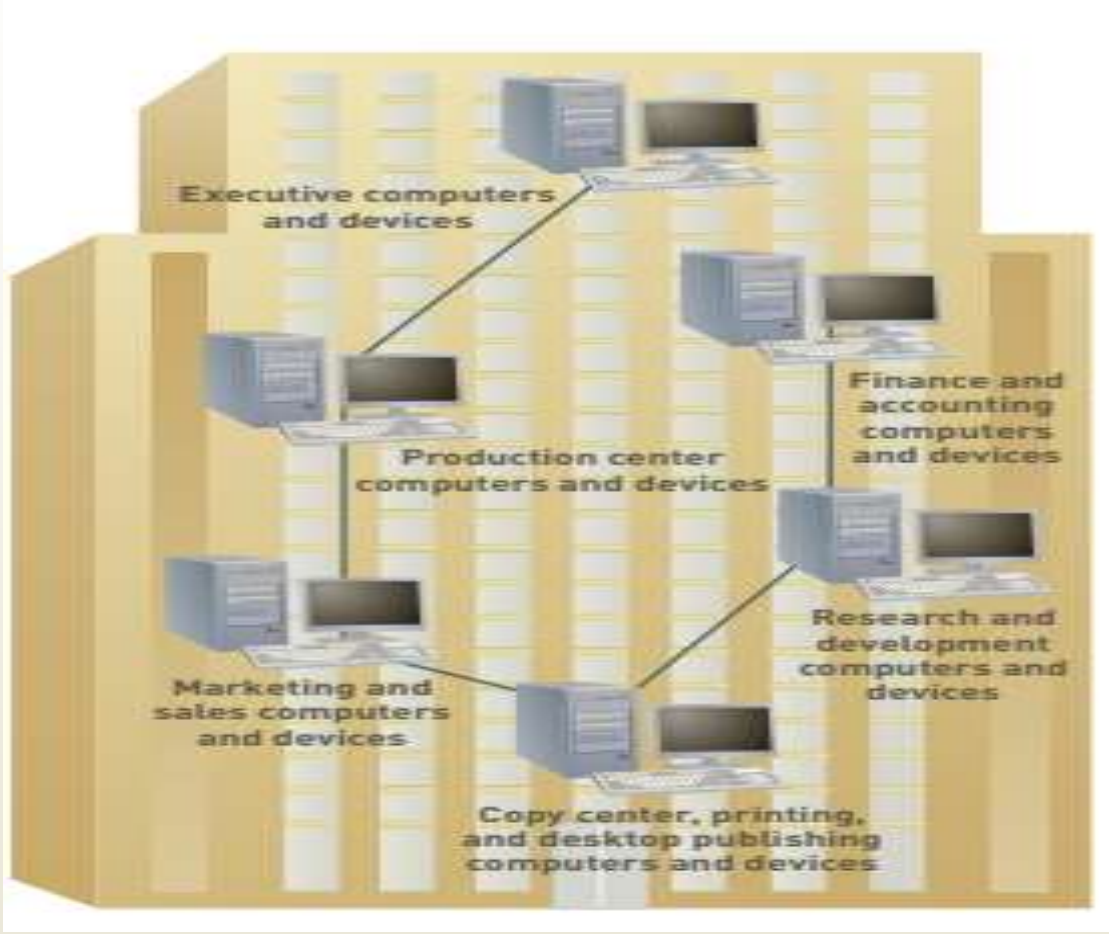


2. Local Area Network (LAN)

- This connects computers and devices within a limited area such as a home, office, or building.
- **Coverage Range:** One room, building, or small campus
- **Key Technologies:** Ethernet, Wi-Fi
- **Real-World Example:** Home or office Wi-Fi network
- **When It Is Used:** For fast and secure communication within a local location
- Usually, a LAN is contained within a single building, but a LAN can ***extend*** to several buildings on a campus.

2. Local Area Network (LAN)

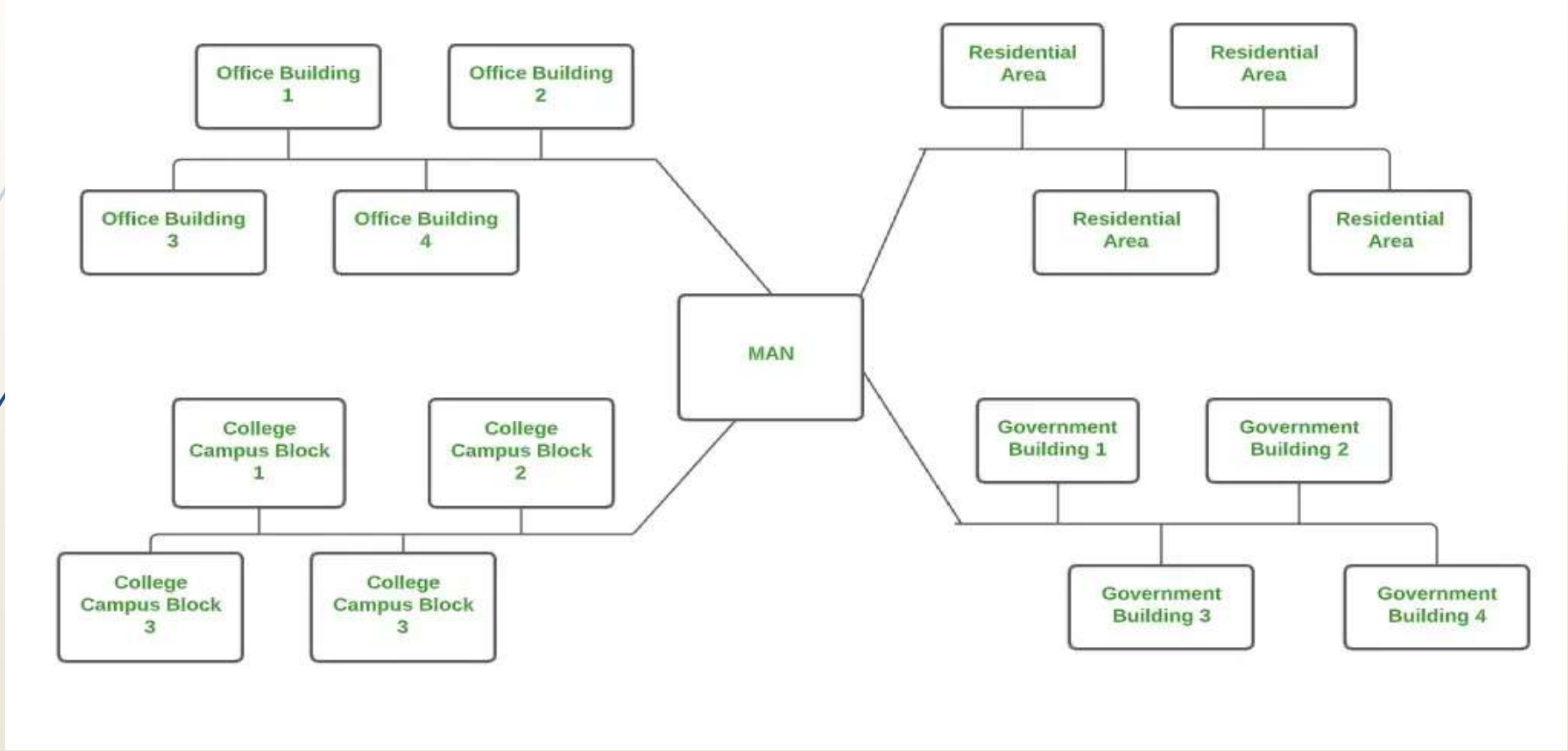




3- Metropolitan Area Network (MAN)

- A MAN connects multiple networks within a city or metropolitan region.
- **Coverage Range:** City or large town
- **Key Technologies:** Fiber Optics, Microwave, Metro Ethernet
- **Real-World Example:** City-wide ISP network
- **When It Is Used:** To provide high-speed connectivity across a city

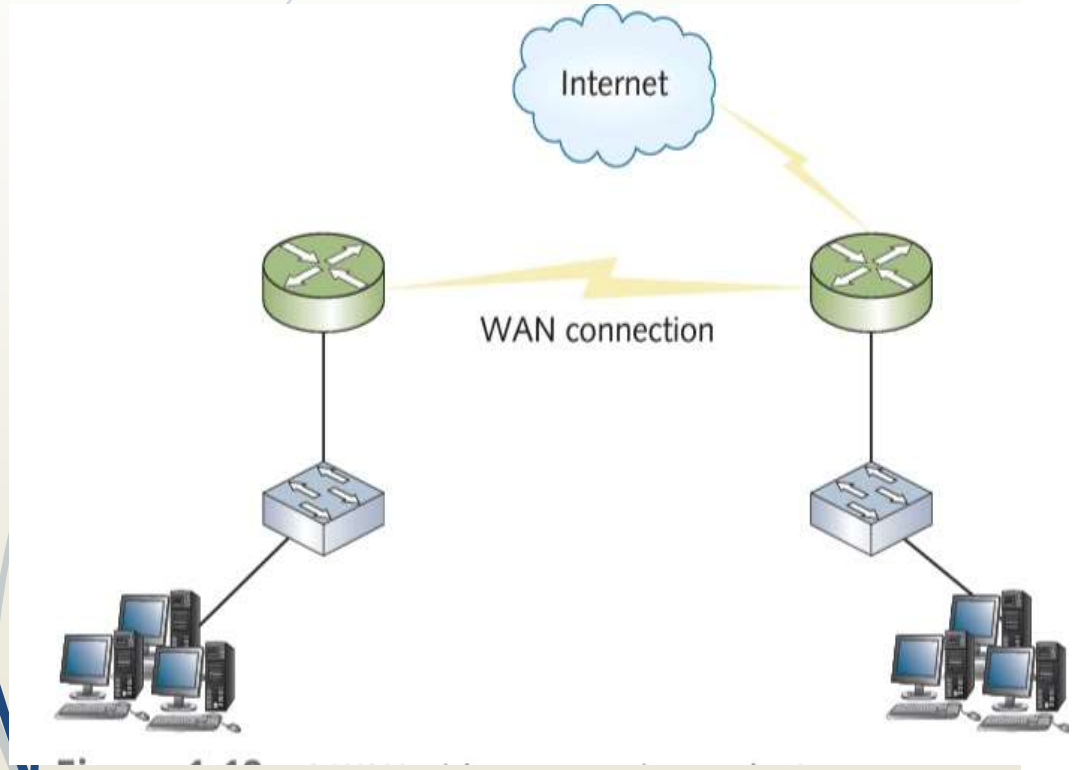
3- Metropolitan Area Network (MAN)

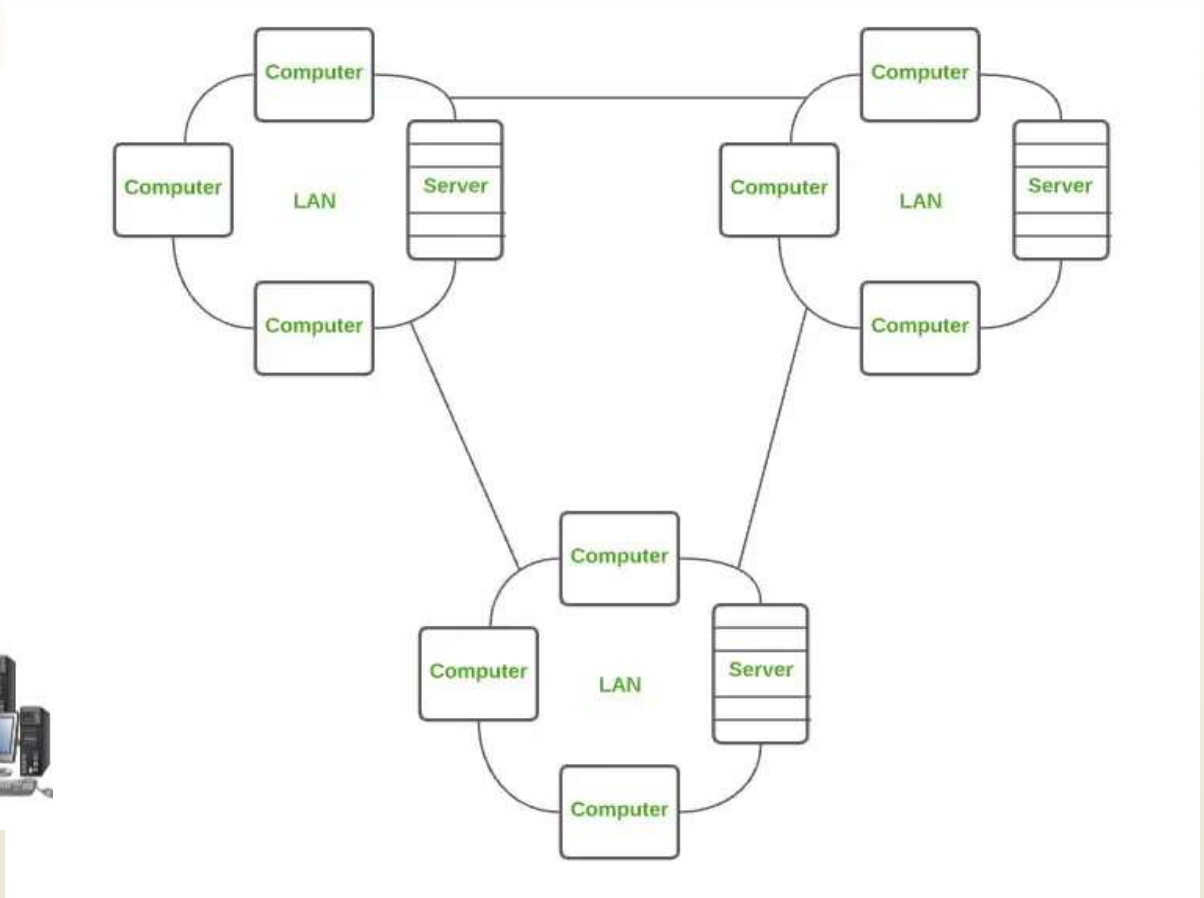


4-Wide Area Network (WAN)

- A WAN connects networks over large geographical areas such as countries or continents.
- **Coverage Range:** Country, continent, or global
- **Key Technologies:** Leased lines, Satellite links, Internet
- **Real-World Example:** The Internet
- **When It Is Used:** To enable long-distance and global communication

4-Wide Area Network (WAN)





Network Topologies

- The shape or structure of a network, including the arrangement of the communication links and hardware devices on the network (**physical topology**).
- The path data travels between computers on a network is considered the network's **logical topology**.
- **Bus**
- **Star**
- **Ring**
- **Mesh**
- **Point-to-Point Topology**

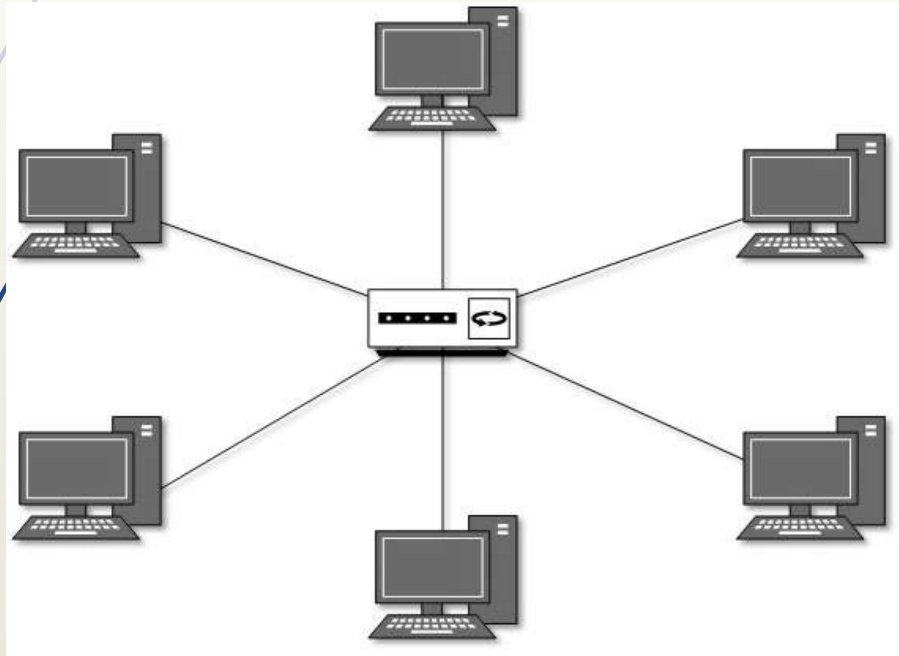
Network Topologies

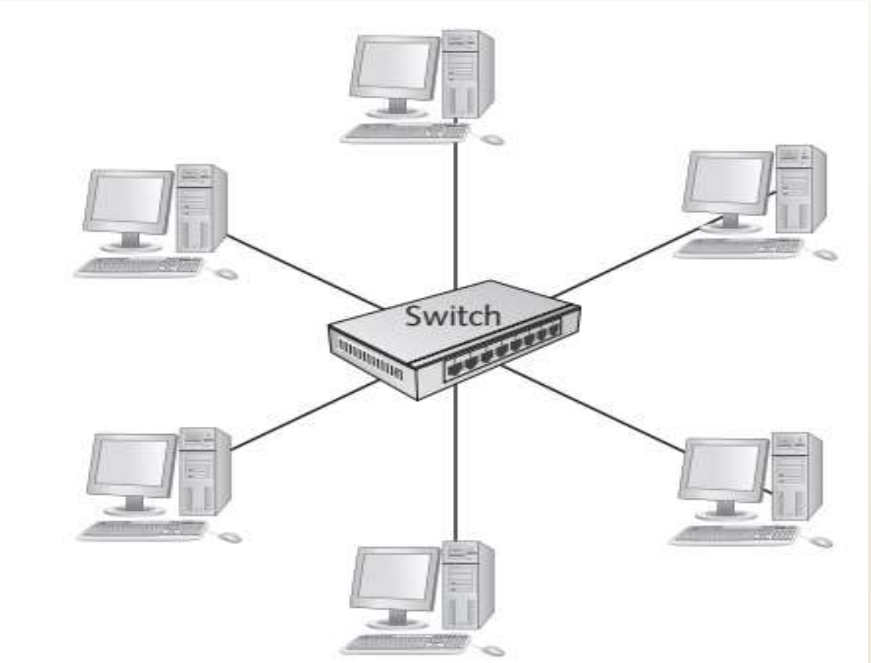
Star Topology:

- A network in which all network devices connect to one another through a single central device called the **hub node**.
- A failure in any link of the star network will isolate only the device connected to that link.
- However, should the hub fail, all devices on the entire network will be unable to communicate.

Network Topologies

- **Star Topology:**





Network Topologies

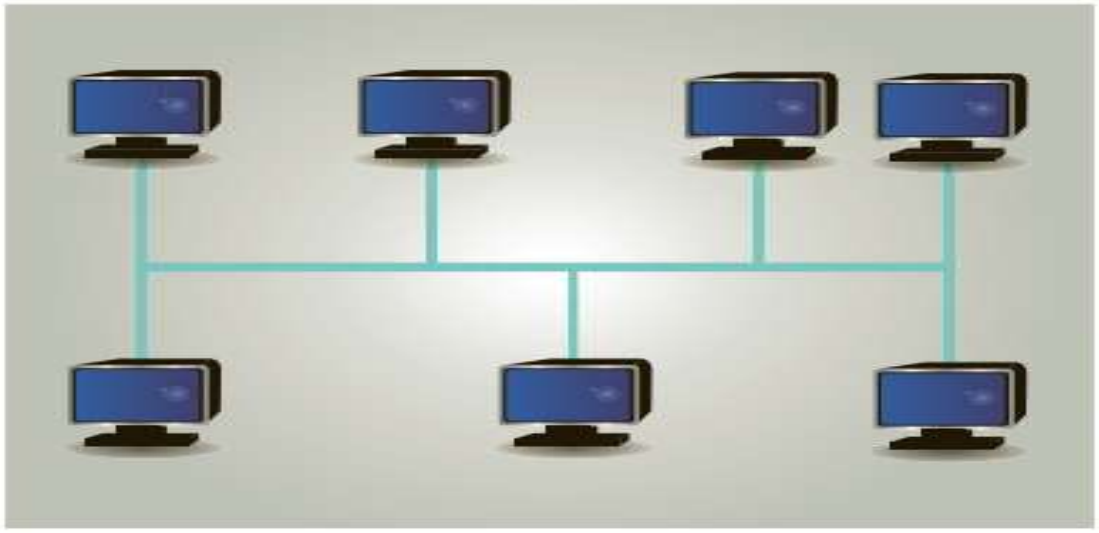
Bus Topology:

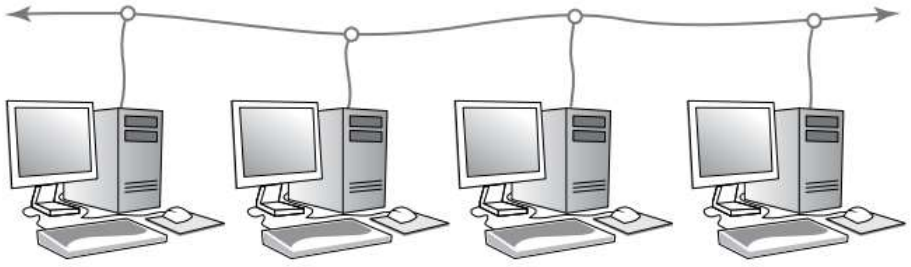
- A network in which all network devices are connected to a common **backbone** (bus) that serves as a shared communications medium.
- To communicate with any other device on the network, a device sends a broadcast message onto the communications medium.
- All devices on the network can **“see”** the message, but only the intended recipient actually accepts and processes the message

Network Topologies

- **Bus Topology:**







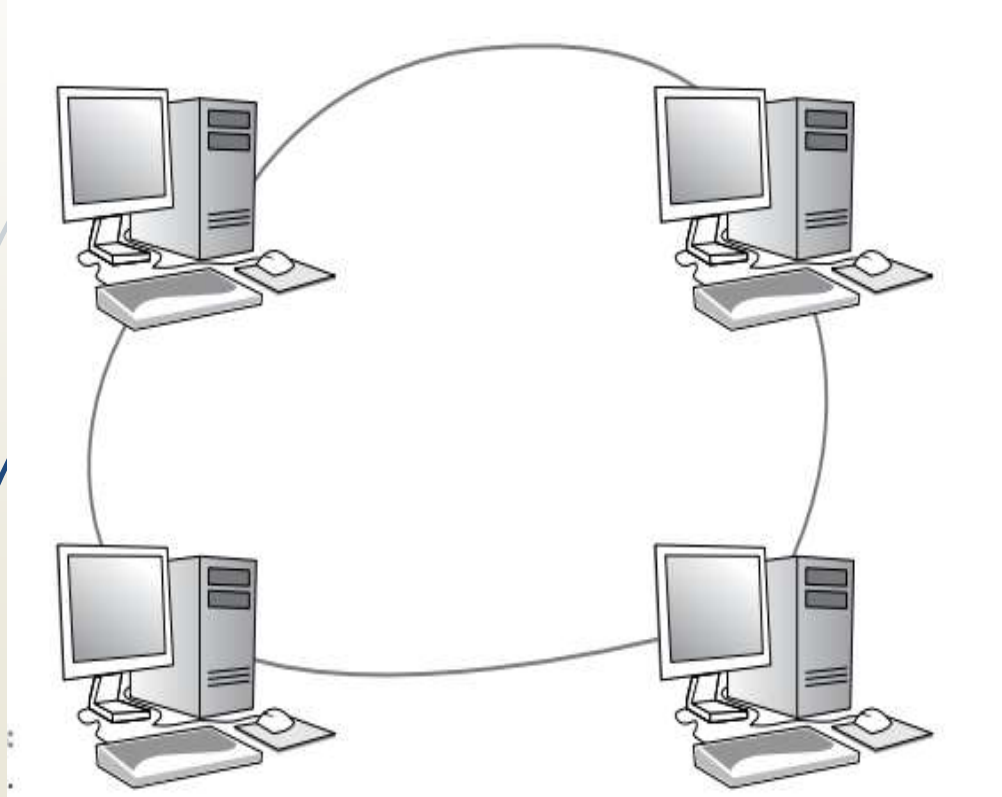
Network Topologies

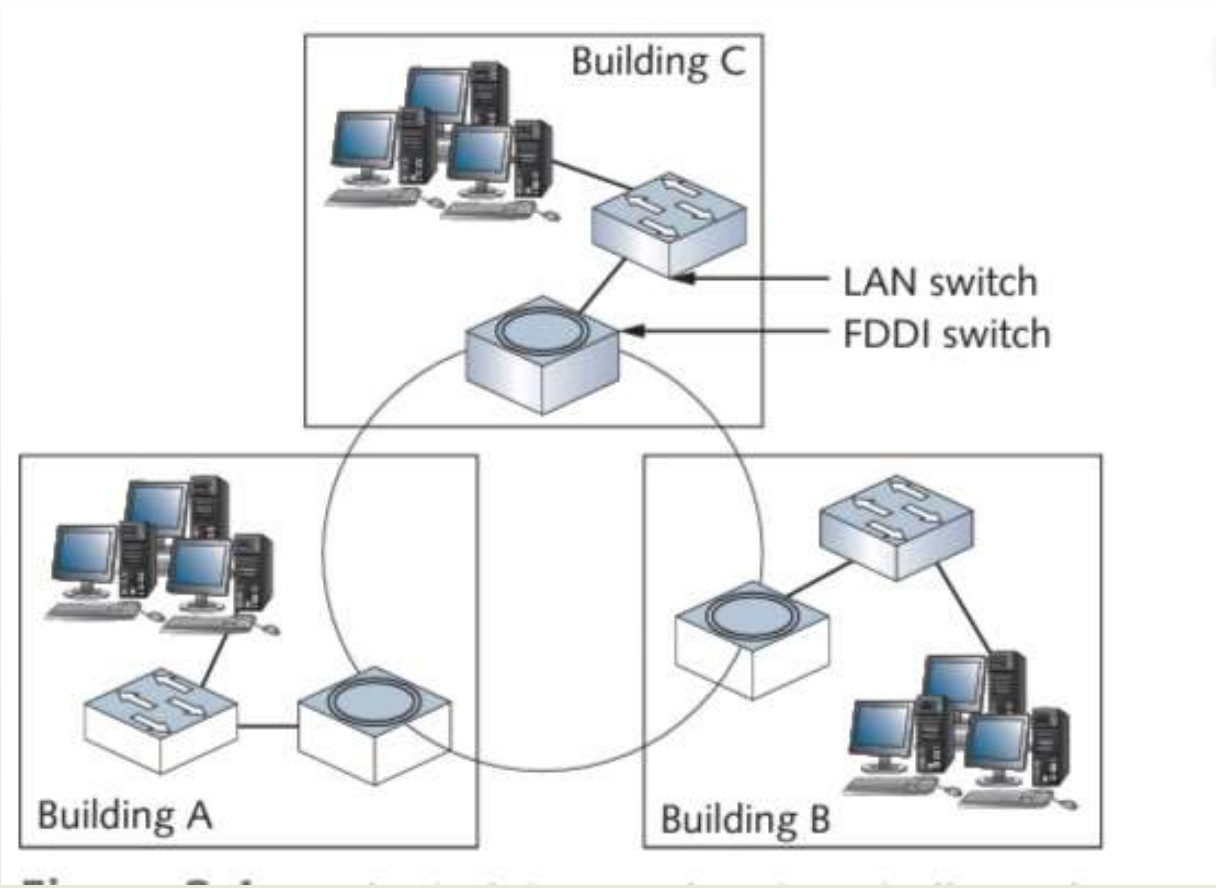
Ring Topology:

- In a ring topology, every device is connected in a circular fashion with each node linked to two other nodes.
- **Packet**: A packet is a message that's sent over the network from one node to another.
- Each computer looks at each packet to decide whether the packet was intended for it. If not, the packet is passed on to the next computer in the ring.

Network Topologies

- **Ring Topology:**





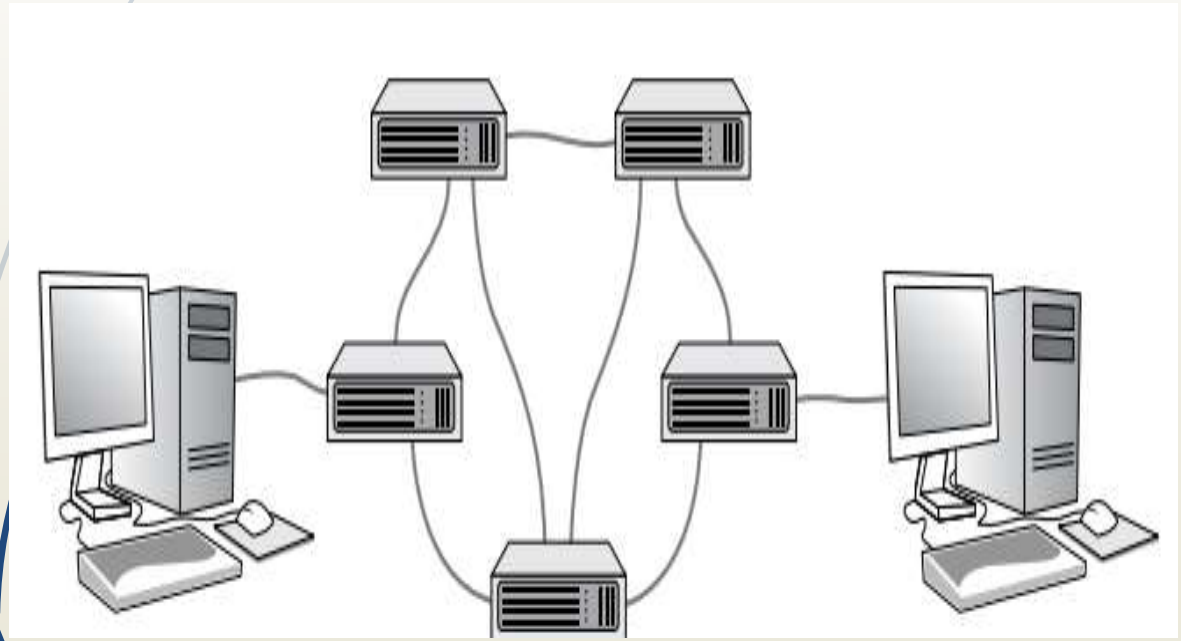
Network Topologies

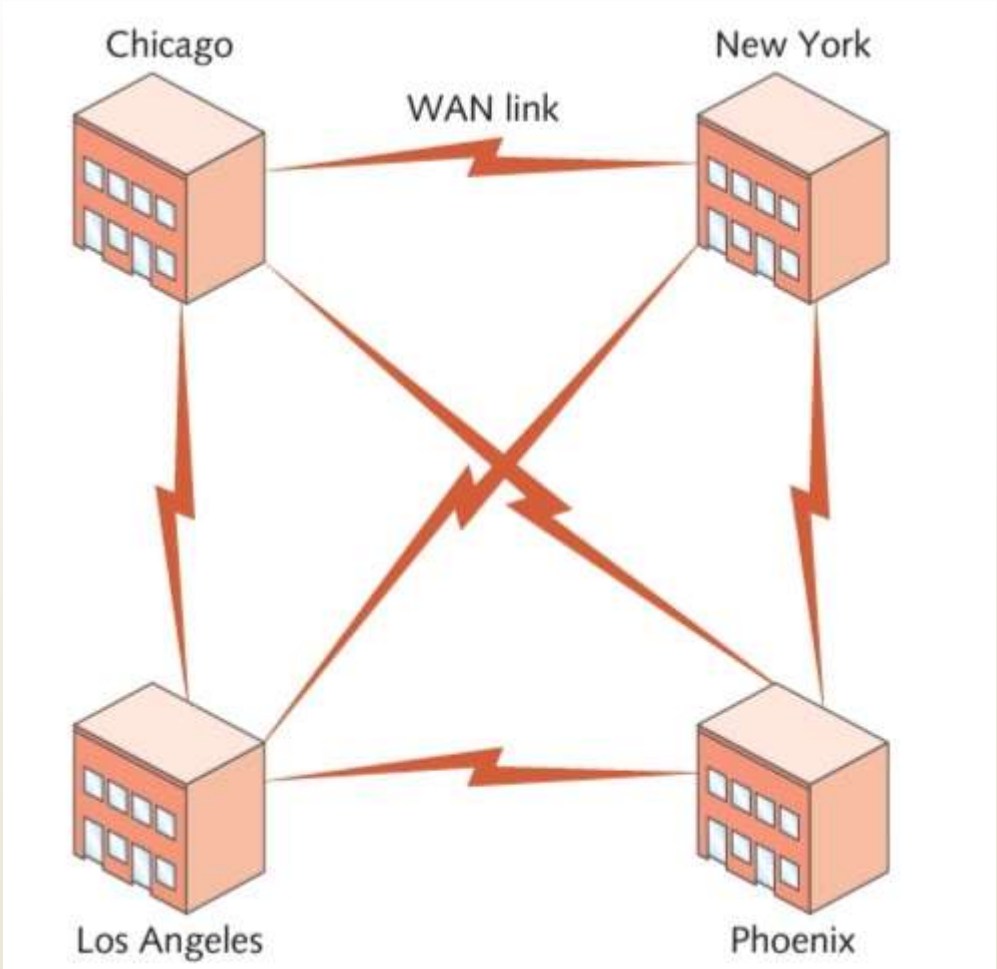
Mesh Topology:

- A mesh topology connects each device to every other device in a network.
- **Fully Mesh:** Every node is connected to every other node.
- **Partially Mesh:** Only some nodes have direct links with each other.

Network Topologies

- **Mesh Topology:**





Point-to-Point Topology

- A **Point-to-Point topology (P2P):** is a direct link between two devices.
- Most often used in **Wide Area Networks (WANs)**.
- Used for *specialized purposes*.



Point-to-Point Topology

- **Common Uses:**
- Connects a business network to a **telecommunication provider**.
- Provides **Internet access**.
- Creates a **WAN or MAN link** to a branch office.
- Used more often in **WANs** and **large internetworks** than in **LANs**.

Point-to-Point Topology

- **Advantages:**
- Data travels on a **dedicated link**.
- **Bandwidth** is not **shared** with other networks.
- **Disadvantages:**
- Can be quite **expensive**, especially for **long-distance WAN links**.
- Allows **only** *two computers* on the network.

Point-to-Point Topology

Other Applications:

- Used in **wireless networks** as a **wireless bridge**.
- Connects *two buildings* without a wired network.
- Extends an existing wireless network.
- Connects *two computers* **directly** for file transfer when a **switch** is unavailable.

Network Devices

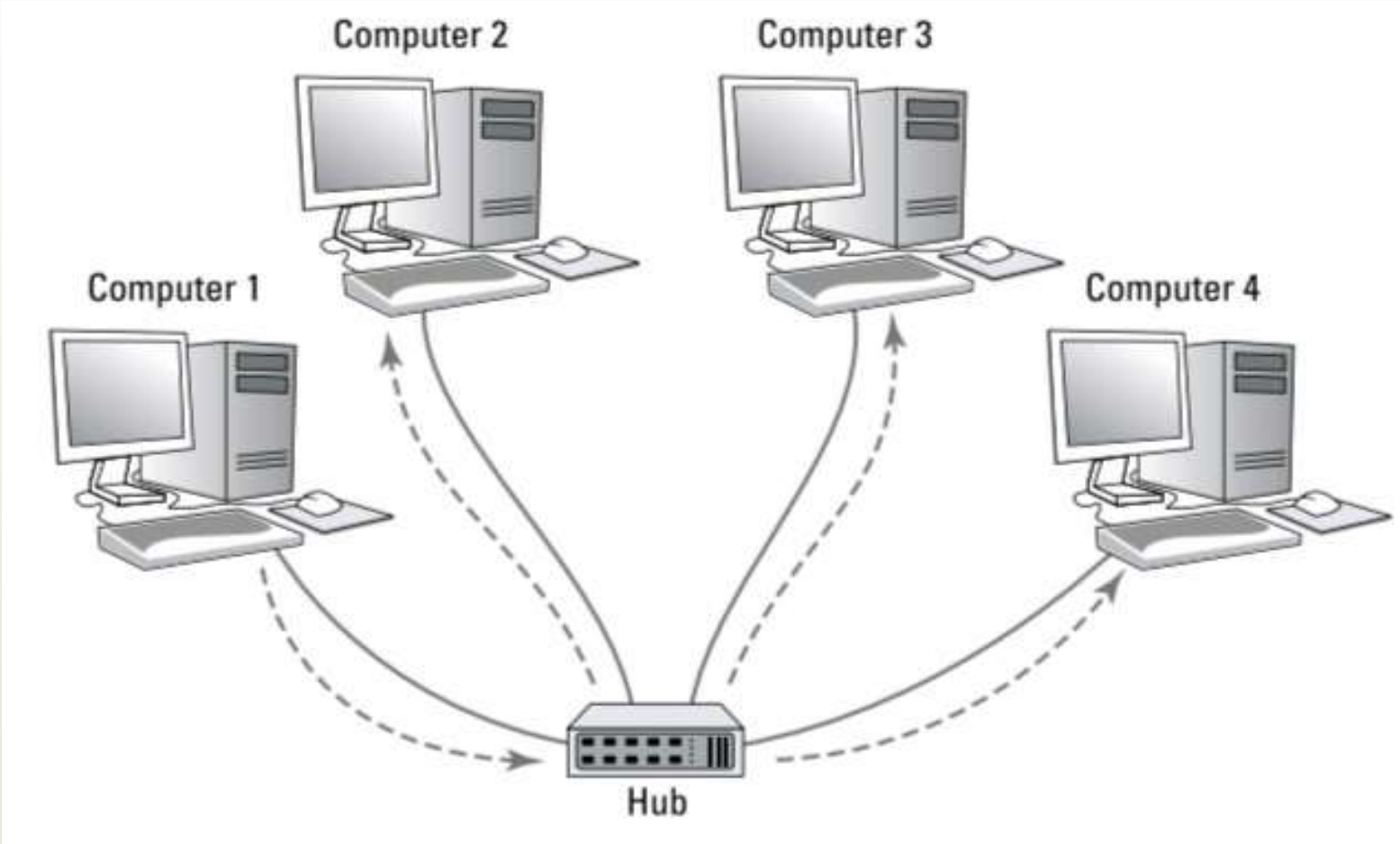
- **1- Network Interface Card**
- NIC provides the physical interface between computer and cabling.
- It prepares data, sends data, and controls the flow of data. It can also receive and translate data into bytes for the CPU to understand.

Network Devices

- **2- Hub:**
- A network hub is a basic physical-layer (Layer 1) networking device that connects multiple Ethernet devices to act as a single network segment.
- It acts as a multi-port repeater, receiving data from one port and broadcasting it to all other ports, leading to inefficient bandwidth usage and shared collision domains.

Network Devices

- **2- Hub:**

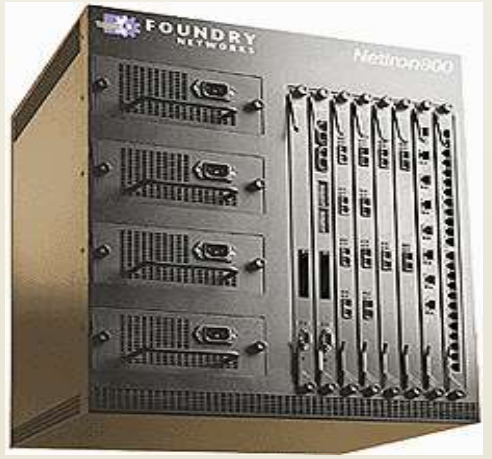


Network Devices

3-Router

- Routers are highly intelligent devices that connect multiple network types and determine the best path for sending data.
- Routers are normally used to connect one LAN to another.
- Typically, when a WAN is set up, there will be at least two routers used.





Network Devices

4- switch

- A switch is a data link layer device.
- A network switch is a computer networking device that connects network segments.
- Network switches are capable of inspecting data packets as they are received, determining the source and destination device of that packet, and forwarding it appropriately.



Thanks !

Introduction to I.T

Lecture 8

Topics:

- The internet & World Wide Web
- 1. The internet
  - 1.1 The internet history
  - 1.2 How the Internet Works
  - 1.3 Accessing the Internet
- 2. The World Wide Web

1. The internet

- The **Internet** is a collection of connected networks.
- In general, these networks are constructed and maintained by organizations called **Internet Service Providers (ISPs)**.

1.1 The internet history

- The ancestor of the Internet was the **ARPANET**, a project started by the U.S. **Department of Defense** (**DoD**) in 1969.
- The goal was to develop the ability to link a variety of computer networks so that they could function as a connected system that would not be disrupted by local disasters.

1.1 The internet history

- Much of this work was sponsored by the U.S. government through the **Defense Advanced Research Projects Agency** (**DARPA**).
- Over the years, the development of the Internet shifted from a government sponsored project to an academic research project.

1.2 How the Internet Works

- In the early days of the Internet, the major communications companies around the world agreed to connect their networks so that users on all the networks could share information over the Internet.
- These large communications companies, called **network service providers (NSPs**), include **Verizon**, **Sprint**, **British Telecom**, and **AT&T**.

1.2 How the Internet Works

- The networks connected through use of the **Internet protocol (IP).**
- **Internet Protocol (IP):** A communication standard that enables computers to route communications traffic from one network to another as needed.
- **Internet backbone**: One of the Internet's high-speed, long-distance communications links (e.g: **fiber-optic cables**).

1.2 How the Internet Works

- The Internet transmits data from one computer (called a **host**) to another.
- The data passed through network in chunks called **packets,** each of which carries the **addresses** of its **sender** and **receiver** along with other technical information.
- The set of rules used to pass packets from one host to another is the **IP protocol.**

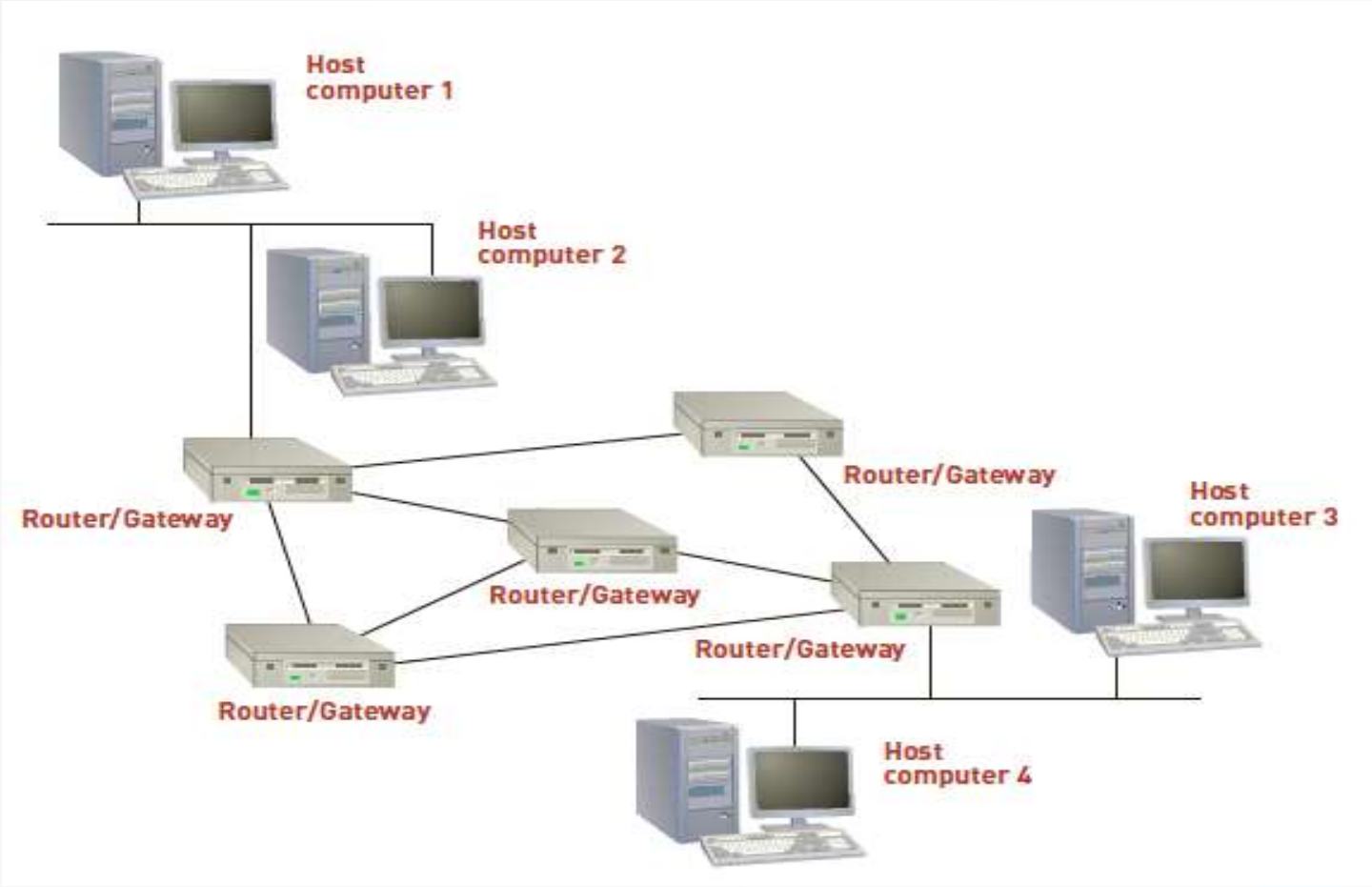
1.2 How the Internet Works

- Many other communications protocols are used in connection with IP.
- The best known is the **Transmission Control Protocol** (**TCP**).
- After a network following these standards links to the Internet's backbone, it becomes part of the world wide Internet community.

1.2 How the Internet Works

- Each computer on the Internet has an assigned address, called its **IP address**, that identifies it on the Internet.
- Because people prefer to work with words rather than numbers, a system called the **Domain Name System (DNS)** was created.
- Domain names such as www.yahoo.com are mapped to IP addresses such as 69.32.133.79 using the DNS.

1.2 How the Internet Works



Routing messages over the Internet

Data is transmitted from one host computer to another on the Internet.

1.3 Accessing the Internet

- **1. Connecting via a LAN Server:**

✓ This approach is used by businesses and organizations that manage a local area network (**LAN**).

✓ By connecting a server on the **LAN** to the Internet using a **router**, all users on the **LAN** are provided access to the Internet.

1.3 Accessing the Internet

- **2. Connecting via Internet Service Providers:**

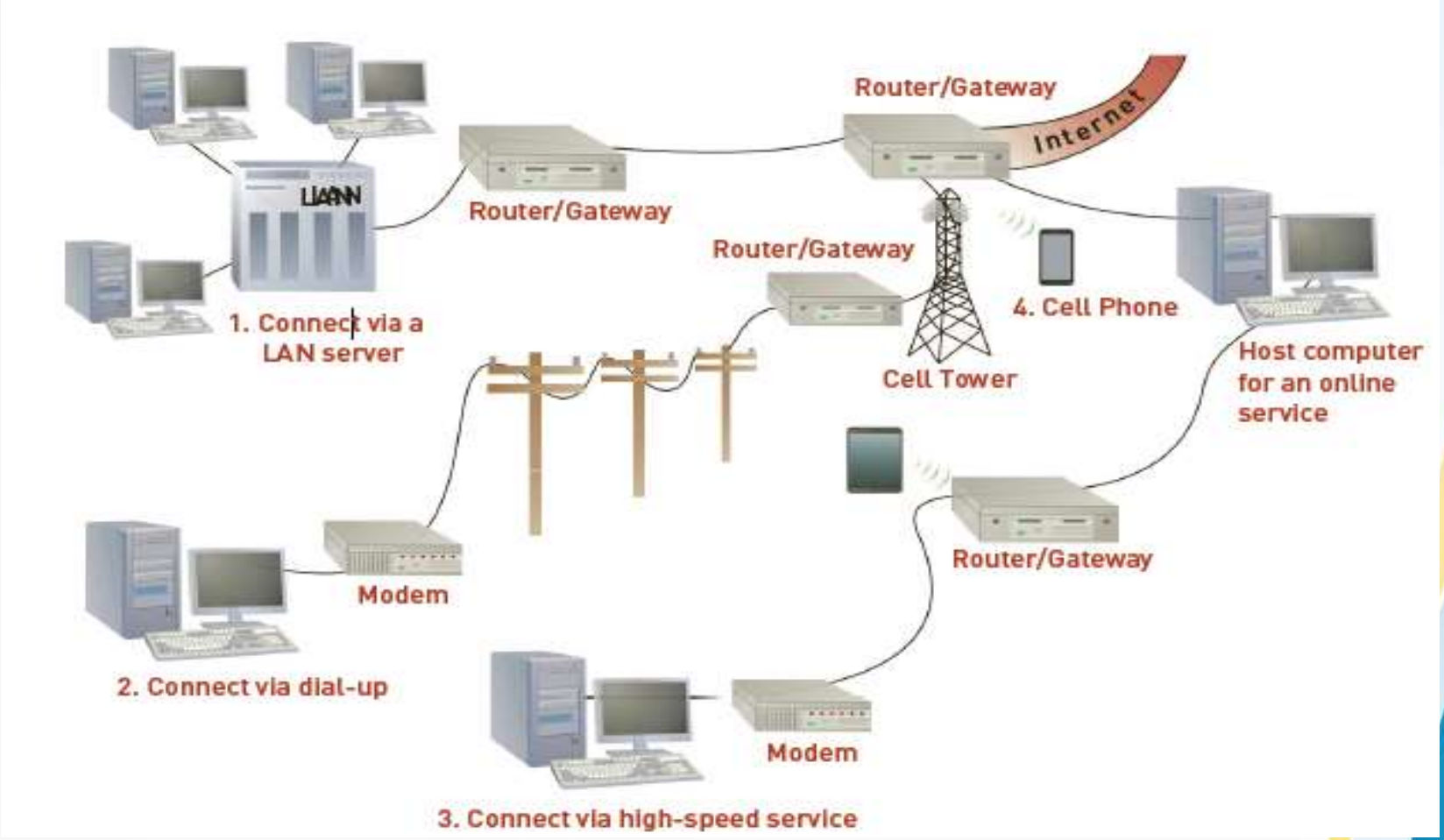
✓ Companies and residences unable to connect directly to the Internet through a **LAN** server must access the Internet through an **Internet service provider**.

1.3 Accessing the Internet

- **3. Wireless Connection:**

✓ Thousands of public **Wi-Fi** services are available in coffee shops, airports, hotels, and elsewhere, where Internet access is provided free, for an hourly rate, or for a monthly subscription fee.

1.3 Accessing the Internet



2. The World Wide Web

- The World Wide Web was developed by **Tim Berners-Lee** at **CERN**, the European Organization for **Nuclear Research** in **Geneva**.
- He realized the potential of combining internet technology with the concept of linked documents, called **hypertext**.
- **Hyperlink**: Highlighted text or graphics in a Web document that, when clicked, opens a new Web page containing related content.

2. The World Wide Web

- **Hypertext** documents are normally transferred between browsers and webservers using a protocol known as the **Hypertext Transfer Protocol (HTTP).**
- A **Uniform Resource Locator (URL)** is a Web address that specifies the exact location of a Web page using letters and words that map to an IP address and a location on the host.
- Example: http://www.google.com.sa

2. The World Wide Web

- The **“www”** part of the address signifies that the address is associated with the **World Wide Web** service.
- Many Internet addresses have a code denoting affiliation categories, such as **com** for business sites and **edu** for education sites.

Affiliation ID	Affiliation
Biz	Business sites
Com	All types of entities including nonprofits, schools, and private individuals
Edu	Post-secondary educational sites
Gov	Government sites
Net	Networking sites
Org	Nonprofit organization sites

2. The World Wide Web

- **Website** is simply data which is stored on www server and which can be freely accessed by people “surfing the web”.
- **Web browser**: Web client software used to view Web pages.
- Some websites can store hidden information about you on your hard disk using **cookies**.

2. The World Wide Web

- **Internet cache**: Each time you display a website within our web browser, a copy of the information (both text and pictures) is saved on your hard disk.
- **FTP** (**File transfer protocol**) is just a way of transferring data from one place to another over the internet.

Questions?